



VinFast

Generic Diagnostic Vehicle Specification VFDSVAEEP0001-V1.5

Specification Book

Scope

Applicable Document of Part 1

Common Electrical Specification for Supplier
Generic Diagnostic Specification for all Vehicle programs.

Range of Validity

Vinfast

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1 General Information

1.1 Language

This specification is written in English.

1.2 Change History

Version	Chapter	Brief Description	Date	Responsible person
1.0	all	Document creation	2021-09-20	Ashad Qureshi Daniel Canelopol
1.1	4.1	Remove table 10	2021-11-15	Ashad Qureshi
1.2	3.2	Update speed signal name Update section 3.2 update extended data with record #12 Section 2.0 update with statement.	2022-01-05	Ashad Qureshi
1.3	6.6.2 Section 3.6, 3.8 3.14 1.2 3.2	Change of soft reset requirement from condition C to M and extended snapshot changed to Record changed sentence. Detail of extended record number clarified LIN specification added Update the requirement of diagnostic pre conditions for general diagnostics Table numbers update and updated table for DTC extended data	2022-01-14	Ashad Qureshi
1.4	3.2	Changed obsolete vehicle speed and speed valid signals table4	2022-02-08	Ashad Qureshi
	3.30	Added a new VIN to cater for Case 2 and text amended	17/03/2022	Ashad Qureshi
1.4	3.2 3.2 3.3 6.3 3.5 3.18	Wording changed and paragraph clarified Vehicle speed status corrected table 5 and removed the motor (and E-motor OFF for BEV) Wait time post ignition increased from 1S to 1.5S and reworded the paragraph and network logging DTC voltage requirements defined ODST test bit 0 definition clarified Fig 6 replaced VOD reference document	08/04/22	Ashad Qureshi



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		defined VOD section clarified		
	1.3	table 2 updated with ISO document		
		DTC mature de-mature strategy clarified.		
	6.0	Title heading number corrected section 6 to 6.35		
	6.26	Table 33 amended to make some sub-functions of service 19 mandatory		
	3.14	Ref of ISO14229 added		
	2.2	Sentence clarified	11/07/2022	
	2.3	Requirement clarified		
	3.4	Requirement clarified		
	3.5	Requirement clarified		
	3.6	Requirement clarified		
	3.8	Requirement clarified		
	3.11	Requirement clarified		
	3.17	Requirement clarified		
	3.21	Requirement clarified		
	3.27	Requirement clarified		
	4.3	Requirement clarified		
	4.5	Table updated with timing		
	4.6	Requirement clarified		
	6.32	Requirement clarified		
	4.8	Note updated		
	3.20	Requirement clarified		
	5.3	Requirement clarified		
	6.2	Requirement clarified		
	6.5	Requirement clarified		
	6.6	Requirement clarified		
	6.7	Requirement clarified		
	6.12	Table 20 updated		
	6.15	Table 23 updated		
	6.28	Requirement Updated		
	6.8	Tables 17, 18 updated		
	6.13	ECU reset service requirement clarified		
	6.31	Service 29 requirement detailed		
	3.2	Table 4,5 updated	12/07/2022	
	4.3	S3 client and server clarified		
	4.6	Functional addressing clarified		
	3.7	Table 9.1 updated		
	3.13	Requirement clarified		
	3.8,3.9, 3.10	Requirement clarified		
	3.14	Added new section 3.14 and table 9.1 moved under section 3.14		
	1.3	Table 2 updated	13/07/2022	
	3.6	Requirement clarified and synced with CAN DB DD01, DD02		
	4.9	Requirement updated	16/08/2022	
	4.3	Table 16 updated	01/09/2022	
	3.4	DTC Strategy defined		



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1.5	4.3	Table 16 updated to include P4 CAN time and St_min	31/09/2022	Ashad Qureshi
	3.2	Table 4 updated	17/10/2022	Ashad Qureshi

Table 1: Change History

1.3 Reference Documents

Reference Number	Source	Title	Version or date	Document Number
1)	ISO	Road vehicles — Diagnostics systems — Diagnostic services	2020	ISO/DIS 14229-1-7
2)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics — Part 7: Data link security	2013	ISO15031-7
3)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics-part 6	2015	ISO15031-6
4)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics — Part 5: Emissions-related diagnostic services	2015	ISO 15031-5
5)	ISO	Road vehicles — Communication between vehicle and external equipment for emissions-related diagnostics — Part 4: External test equipment	2014	ISO 15031-4
6)	ISO	Road vehicles — Diagnostics on Controller Area Networks (CAN)	2021	ISO15765-3
7)	ISO	Road vehicles — Diagnostic communication over Controller Area Network (DoCAN) — Part 5: Specification for an in-vehicle network connected to the diagnostic link connector	2021	ISO15765-5
8)	ISO	Road vehicles — Diagnostic communication over Controller Area Network (DoCAN) — Part 2: Transport protocol and network layer services	2016	ISO15765-2
9)	ISO	Road vehicles — Local Interconnect Network LIN	2016	ISO17987-2
10)	ISO	Road vehicles — Modular vehicle communication interface	2017	ISO22900
11)	ISO	Road vehicles — Open diagnostic data exchange (ODX)	2018	ISO22901
12)	SAE	E/E Diagnostic Test Modes	2015	SAE J1979
13)	SAE	Diagnostic Trouble Code Definitions	2016	SAE J2012
14)	VF	VIN structure	2021	VF-HOMO-DOC-02
15)	VF	VIN Decipher	2021	
16)	CES	CES_Basic_Services_DID_PID_Definition	Latest	VFDSAAEEP0033
17)	CES	CES_Flash_Specification	Latest	CES_VINFAST_EE_VFDSXVEEP0035
18)	CES	CES_Coding_Specification	Latest	See note* VFDSP50000017 (VFDSVAEEP0004)
19)	CES	CES_ECU_Identifier_Specification	Latest	VFDXVEEP0013
20)	CES	CES_Security_Access_Algorithm	Latest	CES_VINFAST_EE_VFDS P50000020
21)	Gov	49 CFR 565 - VEHICLE IDENTIFICATION NUMBER (VIN) REQUIREMENTS*	Check	CFR part 565*
22)	VF	System_Schematics	Latest	
23)	AUTOSAR	Requirements on Diagnostics	Latest	AUTOSAR FO R19-11
24)	VF	CES_ECU LIN_Specification	Latest	CES_VINFAST_EE_VFDS P50000012
25)	ISO	Road vehicles — Environmental conditions and testing for electrical and electronic equipment	Latest	ISO 16750-2:2010(E)
26)				

Table 2 requirements

* Note this VOD is for the current vehicle VFE 8,9.



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The Standard's incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. It is the responsibility of the supplier to check for the latest publications where possible for the development of the ECU if there is uncertainty, then clarifications should be sought from Vinfast or their approved agents acting on their behalf as to which version of the requirement should be used.

*Obtainable from: U.S. Government Publishing Office <https://www.gpo.gov/fdsys/granule/CFR-2011-title49-vol6/CFR-2011-title49-vol6-part565/content-detail.html>

1.4 Document Conventions

In this document, the following terminology applies:

- ▶ "Shall" expresses an obligatory/mandatory requirement.
- ▶ "Should" expresses a recommendation or an advice.
- ▶ "Must" expresses a legal or normative requirement.
- ▶ "Will" expresses a precautionary consideration or an additional/optional feature.
- ▶ "May" expresses a permitted practice/method, not to be considered as a requirement.

2 Introduction

Though an attempt was made to cover all aspects of implementing VF requirements there may still be design attributes that may lack detail or are overlooked. As these and other attributes are further studied, requirements will be added or modified to reflect the latest industry and corporate directions and trends.

2.1 Purpose of Diagnostics

The purpose of the diagnostics can be defined by the following statements: -

- To validate correct assembly
- Check for presence of faults during service or repairs
- To check for communication between the various network nodes that the ECU is connected or communicates through.
- To detect and store a fault regardless of the Off Board Tester making a diagnostic request to obtain this information.
- To report current failures when the On Demand Self-Test is run.
- Legal considerations for diagnostics any Software download solutions that are not via the usual diagnostic communications between the tester and the vehicle shall be reviewed for compliance to right to repair legislation i.e. cannot use proprietary methods to diagnose an ECU.

2.2 Scope

This document covers the basic requirements for Diagnostic task in the current project and describes the main Diagnostic requirements.

The documents:

- Basic DID Definition: - System are mandatory DIDs which need to be supported by all ECU



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suppliers case by case basis as defined in the DID specification.

- Coding Specification: - Defines the requirement if the ECU needs to be coded for adapting ECU behaviour in a vehicle. While applying its general functionality via flash programming
- Flash Specification: - Defines/captures the requirement if the ECU needs to be flashed during production/EOL/After-sales activity
- Security Access Algorithm: - Defines the requirement to implement security for services and functions which need to be protected.

follow the requirements of this specification.

The functionality should be based on the requirements of the Production and

Service. Following types of diagnostic operations are described in this specification:

- Basic Diagnostic functionalities
- Basic ON-Board diagnostic functionality
- Basic OFF-Board diagnostic requirements
- Additional diagnostic features
- Linked specifications and documentations

2.3 Terms and Definitions

OFF-Board communication includes all processes of data communication between a car and the tester device. The **OFF-Board diagnostic** communication is subject to special regulations of OBDII ISO 15031 and, SAE J1979 / SAE J17979-DA. All processes of this communication must be conforming to the requirements of Unified Diagnostic Service (UDS, ISO 14229). Communication over K-Line with OBDII/EOBD relevant System are not allowed in Europe and North America since 2008. This communication must be done over Diagnostic CAN regarding ISO 15031 and ISO 11898.

On-Board communication describes the communication processes between the ECUs over the car communication network.

This includes automatically the **On-Board diagnosis** as an important point which allows self-diagnosis of electronic components and systems. That means that ECUs must be able to detect faults within themselves. On-Board diagnosis makes sure the information and functionality for the Off-Board diagnostic.

A special part of the **On-Board diagnosis** is the fault detection of emissions-related parts regarding OBDII/EOBD. Here the special requirements of the ISO 15031 (part 1 ... part 6) are binding and the requirements of the SAE J1979 / SAE J17979-DA shall be valid.

Note that this part of diagnostic content must be defined by ECU engineer and ECU supplier together, and OEM diagnostic engineer shall be responsible for checking and reviewing, integration testing engineer shall be responsible for validation.

On-Board Diagnostic is required to run in three levels:

- Detection of error/fault
- Conditioning capturing the conditions surrounding the error/fault
- Storing the fault and the data captured during the event in the form of snapshot.

Once a fault is detected it shall be assigned with a unique DTC with the relevant information about the fault must be stored within the ECU to be extracted by the Off-Board Tester during subsequent repairs for analysis and to help in determining the root cause of the fault. The emphasis is therefore on all suppliers to capture as much data as they possibly can during failure event as trying to capture the details surround the event may not be possible.

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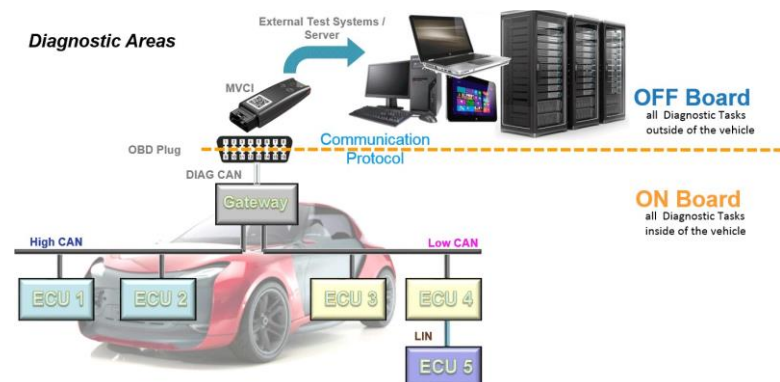


Figure 1: Areas of On and OFF Board

Fig:1

2.4 Abbreviations

DTC	Diagnostic Trouble Code
DID	Data Identifier
PID	Parameter Identifier
RID	Routine Identifier
FC	Flow Control
CAN	Control Area Network
EOBD	European On Board Diagnostic
ISO	International Organization for Standardisation
OBDII	On Board Diagnostic
SAE	Society of Automotive Engineers
KWP	Key Word Protocol
UDS	Unified Diagnostic Services
ODX	Open Diagnostic Data Exchange
SAE	Society of Automotive Engineers
M	Mandatory
U	User conditional
C	Conditional
O	Optional depending on ECU requirement
VIN	Vehicle Identification Number
DOIP	Diagnostic Over Internet Protocol
MFG	Manufacturing
OTA	OTA The

Table 3: Abbreviations

3 Diagnostic Requirements Overview



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3.1 Basic Requirements

The ECUs have to fulfil the requirements of the ISO 14229 UDS. A Gateway routes all necessary diagnostic commands and messages to the connected network components and has to provide the connection and communication to an OFF Board Tester. As diagnostic data exchange format is a complete **ECU ODX template** for every phase shall be delivered to OEM from the suppliers of the ECUs.

3.2 Conditions for Diagnosis

For Diagnostics task on the vehicle, vehicle safety and user safety shall be considered. Therefore, Diagnostic Services required to be supported where ECU either needs software update or where the ECU is requested to carry out a diagnostic task which is behind security whilst the vehicle is in motion some services will not be allowed, when such request is made the ECU shall respond back with NRC \$22 (Conditions Not Correct) unless the following conditions are met for ignition on Stat terminal driver present and vehicle speed is <3km/h. For the details on which services are allowed to respond please see table 4.

Commented [QA1]: Update requirement with

No.	Diagnostic Preconditions	Message ID	CAN Signal	Signal Value	BEV	ICE
1	Ignition ON	0x112	STAT_Terminal	0x02 ON (Driver Present) or 0x04 Software Update OTA	X	X
2	Vehicle Not Moving	0x20D	VehicleSpd and Vehicle SpdSts	Less than 3km/h 0x00 Valid	X	X
Services which are allowed/not allowed to respond when vehicle is not stationary these conditions are to be monitored when exercising these services						
Service	Vehicle stationary IGN ON	Vehicle not Stationary IGN ON	Comments			
10 Diagnostic Session control	Allowed	*Not allowed 02/82/03/83	*Note Session 10 40 only allowed if it is required to have the vehicle running			
11 ECU Reset	Allowed	Not allowed				
14 Clear DTC	Allowed	Not allowed				
19 Read DTC	Allowed	Allowed				
22 Read Data By Identifier	Allowed	Allowed				
2E Write Data By Identifier	Allowed	Not allowed				
2F Input Output Control By Identifier	Allowed	Not allowed				
27 Security Access	Allowed	* Conditional	*Only allowed in session 10 40 if the vehicle needs to unlock the ECU for running routines			
28 Communication Control	*Allowed	Not allowed	*When service 28 disabled communication is disabled no precondition check should be carried out			
31 Routine Control	Allowed	* Conditional	* only allowed In session 10 40 if it is required to have the vehicle running			
34 Request Download	Allowed	Not allowed				
36 Transfer Data	Allowed	Not allowed				
37 Request Transfer Exit	Allowed	Not allowed				
3E Tester Present	Allowed	Allowed				
85 Control DTC Setting	Allowed	Not allowed				
29 Authentication	*Allowed	Not allowed	*Allowed if the service is supported			
2A Read Data By Periodic Identifier	*Allowed	Not allowed	*Allowed if the service is supported			
2C Dynamically Define Data Identifier	*Allowed	Not allowed	*Allowed if the service is supported			
38 File Transfer	*Allowed	Not allowed	*Allowed if the service is supported			
83 Access Timing Parameter	*Allowed	Not allowed	*Allowed if the service is supported			
84 Secured Data Transmission	*Allowed	Not allowed	*Allowed if the service is supported			
86 Response On Event	*Allowed	Not allowed	*Allowed if the service is supported			
87 Link Control	*Allowed	Not allowed	*Allowed if the service is supported			
24 Read Scaling Data By Identifier	*Allowed	Not allowed	*Allowed if the service is supported			

Table 4: Diagnostic precondition



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Note: - If an ECU requires the vehicle to be running to calibrate a sensor for example, then the supplier shall support a special diagnostic session for this activity (i.e. 0x10 40).
If deviation from the above or any other conditions for which Vinfast requires conformity (e.g. for Production/After-sales), the necessary conditions for diagnosis it shall be agreed and documented with the authorised Engineering/ Diagnostic team. Ensuring that correct process is followed, the deviation shall be authorised by designated team members whose names and contact details will be listed in this document or identified in the deviation process. The deviation must be officially listed and captured on a [Vinfast single issue list](#), if this is required on a permanent basis there may be potential downstream customers who may be affected by this change and will need to be informed, else it shall be corrected by the supplier at the earliest opportunity in the program. The reference of the deviation will also be documented in the ODX.

For Flash/Programming tasks and Coding:

All ECU's which support flash programming/coding shall check and fulfil the following preconditions in the pre-programming phase of ECU Flash.

These pre-condition checks are to be reported by the ECU in RID \$0203 response (CheckProgrammingPreConditions), prior to flash programming execution.

Based on RID \$0203 response from the ECU, the diagnostic tester tool will abort the flash programming session, if the required response does not meet precondition for flash. The precondition shall also include reading of the HV via service 22 (DID 016C please see flash specification section 10 onwards)

Basically, a reprogramming or coding process shall only be allowed when the following conditions are met:

No.	Flashing Preconditions	Message ID	CAN Signal	Signal Value	BEV	ICE
1	Ignition ON (and E-motor OFF for BEV)	0x112	STAT_Terminal	0x02 ON (Driver Present) or 0x04 Software Update OTA	X	X
2	Vehicle Not Moving	0x20D	VehicleSpd and Vehicle SpdSts	Less than 3km/h 0x00 Valid	X	X
3	System Voltage Ok	0x10A	*Ubatt	11.5 ~ 16v*	X	X
4	Any other condition which can have a negative impact on electronic components or general functionality regarding vehicle safety i.e. HV off DID 016C					

Table 5: Precondition check



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*These are min and max voltage checks prior to commencing flashing, the ECU shall reject the flash request with conditions not correct if the voltage falls below or above the required threshold.

3.3 Diagnostic Trouble Codes

The ECU shall only log DTCs when Ignition is on and wait 1.5s post ignition as a prerequisite to ensure that vehicle along with all of its other ECUs are under the same normal operating conditions.

Network DTCs shall not be logged below or above voltage as defined by ISO 16750-2 E Ref (25).

The ECUs have to detect fault conditions internally and store Diagnostic Trouble Codes (DTC), if the fault condition is reached (Self-diagnosis). Therefore, internally the ECUs have to check e.g. the input and output for power supply, or the different signals and the CAN network.

If required, in addition the above ECU also have to detect fault conditions which are on the peripheral devices connected to it, i.e. Sensors, Actuators, etc. for which the ECU shall be responsible for.

DTCs are defined by the requirements of the OEM and suppliers working together with input from Development department, Production department and Aftersales department. Any opportunities to correct or enhance diagnostics contents, the supplier has the responsibility of leading this activity which includes to check, review and manage DTC information and its definition.

3.4 Detection of Faults.

- The ECU should be able to identify all of its faults for all components connected on each of its I/O, and external circuits connected to the ECU. The components of the external electrical circuits shall include cables.
- An ECU shall also be able to detect faults other than those in the external circuits connected to the ECU (e.g. all internal faults such as memory failure etc. in the ECU) any failure of the internal component which is not replaceable in service shall be deemed as ECU internal fault and a single DTC for all ECU internal failure type shall be flagged. It may be possible to distinguish different kind of ECU internal failure by associating a different failure byte for it. In the event it is the responsibility of the ECU supplier to capture any the data internally to the ECU surrounding the event which leads/causes the ECU to declare ECU internal fault. It is therefore, advisable for the ECU supplier to carry out any FMEA to identify any externally caused events which may lead to an ECU falsely declaring an ECU internal failure so that these could be identified and corrected by designing them out in the software to prevent from occurring.
- Connectors, sensors and actuators (including both the electrical and the mechanical part of the sensors and actuators). A feasibility study is required and the fault detection shall be implemented wherever a technical solution is available and its cost is justified. The ECU supplier is responsible for the feasibility study. The feasibility study shall be approved by Vinfast or its approved agent of authority acting on their behalf. All detected faults shall be fully specified and documented by the supplier in the ODX with relevant DTCs and help text to assist service individual on the root cause and repair procedures.
- The smallest component replaceable in the workshop shall be identified by one unique DTC. Where possible, each failing part of the component should also be identified by a unique DTC so long as this enhances either the fault tracing capability and/or quality tracking/following-up.



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Example short to ground, short to battery voltage, and open circuit etc. faults should each be identified by a suitable DTC and a unique DTC failure byte describing the error condition. See working example

- 93 80 11 2F (Switch Pack) signal short to ground
- 93 80 12 2F (Switch Pack) signal short to battery
- 93 80 13 2F (Switch Pack) signal open-circuit

The DTC fault detection shall be confirmed using a counter operation for non-emissions related servers shall be implemented as per the details defined in ISO 14229 D.6 this shall be the criteria for logging/flagging DTCs. The incrementation of the counter shall be determined by the supplier of the ECU based on test/monitoring cycle of the signal and therefore the decrement of the fault occurrence counter, where no faults have been detected it shall be half that of the increment counter. e.g. if the fault detection counter is 30 per test failed (i.e. the test time allocated for this this 20ms) where the server passes the test the decrement of this shall be 15 and the time for dematurity shall be 40ms.

All DTCs that are deployed by the suppliers shall have a debounce strategy based on safety use case derived from FMEA or study conducted. All network and missing message related DTCs shall have strategy based on a minimum two consecutive failed counts of detecting a fault based on safety use case.

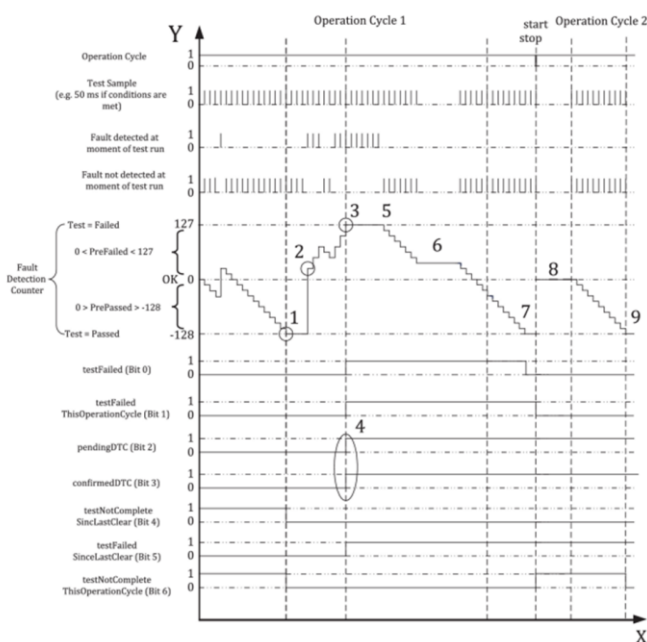


Fig 2



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3.5 On Demand Self-Test (ODST) RID (0303)

The ODST is designed to capture failures as they occur, this routine will therefore be looking at all of the DTCs supported by the ECU and specifically at the DTC status bit such that as soon as the test failed status bit changes 0 to 1 it will flag a problem rather than waiting for the DTC to mature. So if the same DTC was read using service 19 at the same time it may not report an issue if the DTC has not matured sufficiently, with the help of ODST which allows the problem to be detected earlier as the part is being fitted to the vehicle as its being assembled or if this is a part that is being replaced in the aftersales to ensure that it is being fitted correctly. This routine shall be supported by every ECU and will be carried out via a diagnostic routine that shall detect and identify all of the ECU's (I/O circuit faults (*see note)) that could be affected by incorrect assembly or damage to the ECU or its associated components (actuators, sensors etc.) which may be caused by the Vinfast manufacturing process or after sales service fitment/ replacement of the ECU. The routine should use the following guidelines: -

All ECU inputs (e.g., door ajar, 4x4 low ride height etc.) should be physically placed in a defined state and verified to be in that state.

All ECU outputs that employ feedback capability (i.e. all ECU output signals that the ECU can read back) should be energised long enough to detect a fault, and if a failure exists, to identify its type (e.g. open, short to battery/ground, etc.).

All ECU outputs that control subsystem functions that can be monitored (i.e. the ECU can, in some way, read the effect of its control of the subsystem functions) should be actuated and the subsystem's functionality verified (e.g., energising a shock damper motor output that cycles a shock's firm/soft ride type and monitoring a separate shock ride setting input).

Upon completion of the test, the outputs shall return to normal ECU control. An ECU shall complete its on-demand self-test within a maximum time of 30 seconds after receiving the request to perform the test. The routine Control Option Record is not included for this control routine (see reference ISO14229). Any failures that are detected and reported by the ODST shall consist of only three bytes so as to be able to distinguish these from those reported by service 19, and shall be detected when the ("Test Failed bit 0 is set" see table 6) during the control routine execution which means that the DTC will be reported without reaching maturity.

Status Of DTC bit	Bit #	Bit State	Description
Test Failed	0	1	DTC failed at the time of the request
Test Failed This Operation Cycle	1	1	DTC failed during the current operation cycle
Pending DTC	2	0	TC failed on the current or previous operation cycle
Confirmed DTC	3	1	DTC is not confirmed at the time of the request
Test Not Completed since Last Clear	4	1	DTC test was completed since the last code clear
Test Failed Since Last Clear	5	0	DTC test failed at least once since last code clear
Test Not Completed This Operation Cycle	6	0	DTC test completed this operation cycle
Warning Indicator Requested	7	0	Server is requesting warning Indicator to be active

Table 6: DTC status

see example below: -

DTC reported using service 19; 0x94 55 11 1B.

Same DTC reported using service 31 (03 03) 0x94 55 11.

*note: - Omitting the evaluation of an ECU I/O circuit shall only be permitted for the following reasons: -



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- Testing a circuit would exceed the 30-second on-demand self-test execution time.
- Testing a circuit would require operator interaction during the test.
- Testing a circuit could damage the ECU and/or any of its components.
- Testing the circuit could cause a hazardous ECU/vehicle condition.

In order for an ECU to qualify for an exemption the supplier will need to show a use case where evidence gathered by FMEA or risk analysis has shown that running of the routine is likely to lead to an event defined above.

3.6 Snapshot data record

Global/local snapshot information includes all of the following as defined in table 7, 8: -

This Global snapshot record number is equal to 0x10 for the current DTC status, however ECU can also support DTC record number 11 thereby if the failure were to reoccur the two snapshots could be compared to identify if there are any changes in the environment or condition etc. All snapshot DIDs shall be left justified and any remaining data field which remains empty shall be padded with zeros. DTC Snapshot Record Number shall be supported for all numbers valid for the ECU and for

0xFF (meaning report all records). Snapshot records reported by this service shall be reported in single string and in the following order with time stamp followed by Odometer reading then vehicle voltage etc. see working example below.

Signal name	Cvt	Start byte	Description	DID length in Bits
Time stamp	M	#1	0x F120	48 as defined below
			Date of "Year"	8
			Date of "Month"	8
			Date of "Day"	8
			Date of "Hour"	8
			Date of "minute"	8
			Date of "second"	8
Odometer reading	M	#2	0x F121	24
			odometer	
Supply voltage	M	#3	0x F122	14
			Vehicle Supply voltage	
Ignition status	M	#4	0x F123	8
			Ignition status	
Vehicle Temperature	M/O	#5	DD01* *Mandatory based ECU location	8
			Cabin temperature	
Outside Temperature	M/O	#6	DD02* Mandatory based ECU location	8
			Outside Temperature *Mandatory based ECU location	

Table 7: Global snapshot data record



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Working example of Snapshot record: -

Where DD03, DD04 is a local snapshot of the DTC.

19 04 92 8F 11 FF

59 04 92 8F 11 2A 10 08 F1 20 21 10 15 20 30 59 F1 21 00 00 01 00 F1 22 91 F1 23 02 DD 01 19

DD 02 19 DD 03 21 DD 04 78

Local Snapshot record number is equal to = 0x09 to allow for additional local snapshots to be added

Signal name	Cvt	Start byte	Description	Bit length
ECU Temperature	M	#7	0xDD03	8
			ECU temperature	
ECU supply Voltage	M	#8	0xDD04	8
			ECU voltage	

Table 8: Local snapshot data record

Note. The content of table 7,8 shall serve as a bare minimum support for snapshot data but ECU suppliers are encouraged to take a snapshot of whatsoever they deem necessary to assist them in root-causing the fault/issue. If the ECU has the hardware required to take and report ECU voltage/temperature, then its mandatory for the ECU to report this in the local snapshot, if the ECU does not have the necessary hardware then it should report the global vehicle voltage and temperature.

DD01 Vehicle Cabin Temperature

CAN signal CCU_InteriorTemp

Length = 8 Bits

Unit = Degree C

Value Type = Unsigned

Initial Value = 0

Factor = 0.5

Offset = -40

Min value = -40

Max value = +85

DD02 Outside Temperature

CAN signal CCU_TempEx_AC

Length = 8 Bits

Unit = Degree C

Value Type = Unsigned

Initial Value = 0

Factor = 0.5

Offset = -40

Min value = -40

Max value = +85

Please Note: -In the absence of the temperature/voltage sensors which are missing from hardware design, the supplier must show through FMEA or a feasibility study how they will protect the device. If in the unlikely event the internal conditions of temperature/voltage or any other conditions were to



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increase beyond the max or the safe operating range of the device, since operating a device beyond its minimum or maximum operating range is likely to cause stress of the internal components and therefore result in shortening the service life of the ECU.

Time stamp: Dates are always transmitted as Unsigned codes in 6 bytes. The order of the numbers is 'yymmddhhmmss'. The detailed information can be found in the CAN matrix. If the signal is not available from the CAN bus, the maximum value 0x FF FF FF FF FF FF shall be stored.

Odometer reading: The odometer reading will be transmitted as 3 bytes with a resolution of 1 km/bit. The valid range will be from 0 to 16777214. After the maximum valid value (16777214) is reached, the counter will stop and stay at this value. If the bus communication has failed or an invalid odometer value is received, the ECU shall use the last known valid odometer value for the current power cycle, based on CAN spec.

Voltage: Voltages are transmitted as 14 bits with a resolution of 0.0009777 V/bit. This leads to a valid value range of 0 -18V. The maximum value 0x FFFF indicate the signal is not available.

Ignition status: The definition of ignition status signal is in the following:

Signal name: **Ignition status**, Data of length: **3bit**, Value table is as follows:

0x00=Off
0x01=ACC
0x02=On
0x03=Start
0x04~0x06=reserved
0x07=invalid

The DTCs have to be stored as **3 Byte Code** and its status for the UDS components including powertrain and chassis ECUs. Therefore, the Supplier of the ECUs have to follow the requirements of the ISO 15031, ISO 14229 and actual SAE J2012.

Any ECU should have sufficient memory to store as minimum 20 DTCs in a non-volatile memory. The memory must be big enough to store the DTCs inclusive their defined freeze frame data and Local snapshot.

Addition to the above global snapshot information, some other useful information for the error detection is taking a ECU local snapshot.

The requirement of Storage of DTC local snapshot data is specified by the supplier and is specific to each DTC. This data Shall be chosen so that it will be useful in identification of the conditions in which the fault occurred (e.g., operation state and surrounding environment of the ECU) which are present whenever a fault or an intermittent fault (and the fault symptom) occurs.

Please note: - The substituted or default values should not be used for snapshot data as they do not add value or provide any useful information.

3.7 Extended data

Extended data information is a set of data which provides extended status information associated with a DTC; it contains fault occurrence counter, aged counter, aging counter and priority. The bytes required for extended data record and the byte sequence supported in the "DTC Extended Data Records" of response message are defined in following tables.

This extended record data type number is equal to 0x11 which shall capture and report the content of table number 9 as a string proceeding the record number 10, see section 3.13 for a working example of this data record. *Note. The content of table 9, serve as a bare minimum support for extended data record but ECU suppliers are encouraged to maximise on the reporting and gathering of information to be useful when trying to find the root cause of the failure. The content table 9.1 is optional.



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Extended data record type 11	Fault occurrence counter 01	Aged counter 02	Aging counter 03	Priority 04
Convention value (Cvt)	M	M	M	M
Data length	1 Byte	1 Byte	1 Byte	4 bit
Byte position	Byte 0	Byte 1	Byte 2	Byte3(bit0~bit3)

Table 9: Global extended data record



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3.8 Fault Occurrence counter

The fault occurrence counter indicates how often an error has occurred since the drive cycle has started or last successful DTC clearing was carried out

This parameter shall be stored in non-volatile memory. Fault occurrence counter will be incremented by 1 when the “test Failed This Operation Cycle” bit is set to “1” from “0”. When this number overflows, it would always equal to 254 and can’t be changed if a new error is occurring. The format of occurrence counter is defined in the following table: 10:

Data name	Data length	Data type	Resolution	Initial value	Overflow	Fault value/not available value
Frequency of occurrence	1 byte	Unsigned Integer	1	0	254	255

Table 10 the format of occurrence counter

3.9 DTC Ageing management

The deletion of a DTC shall only be possible with diagnostic service if the DTC i.e. not a permanent fault. Also all related information except aged counter can be deleted autonomously after regular conditions are retrieved, optional combined with a defined amount of ignition cycles or a defined period of time. System designer should make decision on whether a DTC can be deleted autonomously or not. The number of ignition cycle should be counted by ECUs themselves via clamp signal. Note See section D7 and D11 of ISO 14229 requirement

3.10 DTC ageing criteria

A process whereby certain servers evaluate past results of each internal diagnostic to determine if a confirmed DTC can be cleared from long-term memory, e.g. in the event of a calibrated number of failure free cycles. Every DTC which could be aged shall be specified a number as DTC aging criteria, if during the previous “DTC aging criteria” times consecutive ignition cycles, the test ran and passed in each ignition cycle, a particular DTC will be allowed to be aged. If no other stated in the DTC list, the value “DTC aging criteria” this shall be equal to 40dec



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3.11 Aging counter

This example provides an overview on the operation of a DTC Aging Counter which counts number of driving cycles since the fault was latest failed. (see ISO14229 D.7)

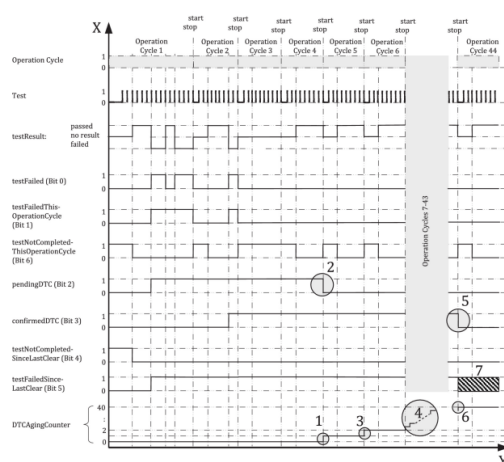


Figure 3 specifies the DTC Aging Counter example.

- DTC Aging Counter is incremented after completing an operation cycle in which test did not fail. pendingDTC is set to zero after an operation cycle in which the test completed and did not fail. In case an ECU does not support a power down sequence (i.e. is immediately shut off when the ignition is turned off) it will be unable to detect the end of the operation cycle. Therefore, it is also valid to set the pending DTC bit to zero at the beginning of the next operation cycle.
- DTC aging counter, counts number of driving cycles since the fault was latest failed excluding the driving cycles in which the test has not reported "testPassed" or "testFailed".
- DTC Aging Counter continues to increment because test is not failing during these operation cycles confirmed DTC is set to zero when aging criteria is fully satisfied (e.g. DTC Aging Counter reaches a specific value).
- DTC Aging Counter reaches a maximum value (e.g. 40) at which time the confirmed DTC bit is cleared it is the responsibility of the supplier to specify whether or not Test Failed Since Last Clear bit is reset by aging-criteria or due to an overflow of the fault memory.

Specific counters for OBD (e.g. number of remaining driving cycles until the "check engine" lamp is switched off if driving cycle can be performed in a fault free mode);
If the DTC can't be deleted autonomously, ECU must save the aging counter status 255dec in the error memory.



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Data name	Data length	Range	Fault/invalid value
Aging counter	8bit	0<agingCounter<DTCAgingCruteria	255

Table 11: aging counter definition

For the aging counter mechanism can refer to (ISO 14229).

3.12 DTC aged

This is a signal which indicates that a DTC has aged sufficiently to be removed from the ECU. If the DTC can't be deleted autonomously, ECU must save the aged counter status 255dec in the error memory.

Data name	Data length	Range	Fault/invalid value
Aged counter	8bit	0...254	255

Table 12: aged counter definition

3.13 Fault warning lamp extinguishing process

DTC with lamp indicator shall turn off the fault warning lamp, as part of the DTC aging management.

For non-emission-related DTC with lamp indicator, if during the previous 3 consecutive valid ignition cycles, the particular fault warning lamp will be allowed to be extinguished, and the "warningIndicator-requested" bit of DTC status byte shall be cleared before it.

Working example of Extended Data being reported: -

19 06 92 8F 11 FF

59 06 92 8F 11 01 01 02 01 03 00 04 01 10 7F 11 = Record type 2 xx xx xx x

See explanation below: -

01 (DTC Extended Data Record Number) 01 (Extended data record value for 01) = "Record counter 0x01 Cycles since last detected".

02 (Extended data record number) 01 (Extended data record value for Record 02) = "Record counter 0x02 Cycles since last detected in which test completed".

03 (Extended data record number) 00 (Extended data record value for Record 03) = "Record counter 0x03 Cycles since first detected".

04 (Extended data record number) 01 (Extended data for 04) = "Record counter 0x04 Number of cycles in which fault detected".

10 xx (See Fig 2)

11 = Record type 2 xx xx xx x (see table 9).

Note: Reference [Ref. 1 ISO 14229] has re-defined the use of this service since the 2006 version of the standard. Previously the standard was not clear about whether DTC Stored Data Record reported the same data as DTC Snapshot Record, but it is now made clear that the data is different. DTC snapshot Data stores data for each DTC from the point of view of the DTC, including the event that caused the storage. i.e. it relates to storage areas that are created dynamically for each DTC fault event. DTC Stored Data stores data related to a fault event from the point of view of the data storage area, including the DTC that relates to the storage. i.e. it relates to a fixed series of storage areas that can be filled with data

The DTC Extended Data records shall be reported according to reference [Ref. 1] and shall not be



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reported with any service other than service 0x19 Sub-functions x06. Note: reporting with service 0x19 sub-functions other than 0x06 is optional. DTC extended data must always be reported with its associated DTC. The extended data has no meaning without its related DTC.

* see table 4

3.14 DTC Extended Data additional explanation

DTC Extended Data Record number 2	Cvt	Size in bytes	Type	Value and data description	Description
*01	U	1	INT	0-255 count	Record counter 0x01 Cycles since last detected
*02	U	1	INT	0-255 count	Record counter 0x02 Cycles since last detected in which test completed
*03	U	1	INT	0-255 count	Record counter 0x03 Cycles since first detected
*04	U	1	INT	0-255 count	Record counter 0x04 Number of cycles in which fault detected
*10	U	1	INT	-128- +127	DTC Fault Detection Counter

Table 9.1 DTC Extended data records.

These counters are more intimate than those given in Table 9 where a sequence of operation is fundamental to diagnosing a pattern for DTC fault detection and prediction.

Sequence at the onset (occurrence) of a fault i.e. when fault is first seen by the ECU the counter No 4 starts to count the Cycles in which the fault was present If the fault is present in the following cycle the counter 3 will start to increment along with and following counter 4 If the fault disappears by its own accord then the counter 3 will continue counting and counter 4 will remain static If the DTCs are cleared by the tester all counters are reset.

If then the DTC returns counters 1 & 2 start to count if the fault has completed all action to determine if the fault is Confirmed, then counter 2 will count else counter 1 will count also counter 4 and 3 will count based on ECU drive cycle conditions.

Counter 0x10 basically will read and display the current status of the DTC based on FIG 2. And give a value between +127 = Test Failed 0 = Test OK -128 Test Passed (This also depends on DTC step calibration.)

3.15 DTC clearing

A DTC can be cleared in two conditions, a diagnostics service request has been sent for a DTC clear or a DTC has aged.

When a DTC is cleared via a clear DTCs diagnostics command, all information connected to that DTC will be reset, i.e. the following parameters will be set to "0":

Fault occurrence counter

Fault pending counter

Ageing counter

Aged counter

DTC status byte

Global snapshot data

Local snapshot data

Note: - if a DTC clear is used to clear a Hard fault using service 14 all of the above counters will be reset but the DTC will still be active.

When a DTC is cleared via the DTC ageing mechanism, the same parameters except for the aged counter will be cleared. The aged counter is retained in ECU long term memory to allow



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the rectifier to determine if that particular fault was previously deleted due to the ageing mechanism.

3.16 Diagnostic Jobs

If required, the ECUs have to support Services for read and write data and parameter. Logistic data, e.g. HW number or SW number of a ECU, should be supported by all ECUs in the same data format. Production and Service have to define their special requirements together with logistic department and diagnostic department.

3.17 Routines for Actuators

All ECUs which are able to actuate internal functionalities with a use of a routine e.g. actuate relays or lamps, etc. should implement such routines so that these functionalities of these ECUs could be checked during Production and Service, the suppliers have to define these and document how to check such functionalities.

3.18 Variant Coding

Relevant ECUs (see component specifications) should be code-able by software and support on this way different versions or functionalities of the ECUs can be actioned. VOD will be a subject of separate document which will capture the required input from ECU owner, suppliers, marketing and other stakeholder about what the features are in the current scope and what features are intended to be proposed for the future and how these are intended to be used in VOD and how should the values should be configured for adapting ECU behaviour based on market the vehicle will be sold in and also enable the control of any subscription based feature options.



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3.19 Flash / Reprogramming

Relevant ECUs (see component specifications) should support flash functionality. This functionality is needed for a fast and easy change of software during the development process, in Production and in Service.

3.20 OBDII / EOBD requirements

In case of OBDII/ EOBD relevant tasks, the relevant ECUs has to support this regarding the actual specification ISO 15031 and SAEJ1979 / SAE J1979-DA. Following services must be implemented in the OBD relevant ECUs regarding the requirements of the ISO.

OBD Service	Service Name	Usage
\$01	Request current powertrain diagnostic data	M
\$02	Request powertrain freeze frame data	M
\$03	Request emission-related diagnostic trouble codes	M
\$04	Clear/reset emission-related diagnostic information	M
\$05	Request oxygen sensor monitoring test results	N / A
\$06	Request on-board monitoring test results for specific monitored systems	M
\$07	Request emission-related diagnostic trouble codes detected during current or last completed driving cycle	M
\$08	Request control of on-board system, test or component	M
\$09	Request vehicle information	M
\$0A	Request emission-related diagnostic trouble codes with permanent status	M

Table 13: OBD Services

3.21 Customer specific Adjustments

The ECUs support additional requirements, based on the specification from Production Team and Service Team. Follow requirements exist:

In order to support the production environment, the control units should support 'production' Mode. All service modes which are necessary to perform the production process must be accessible from this mode.

The supplier has to provide a '**production proposal**' description for the control unit in a separate component diagnostic document, which must contain all necessary steps of the production process.

The following tasks are mandatory and need to be explained in details:

1. methods and routines which are required to initialize the control unit
2. adjustment or calibration like around view monitor system
3. Filling and evacuation
4. Variant Coding
5. MHU FOTA/SOTA implementation Process Procedure
6. OBD related tests
7. Learning process like between key and immobilizer
8. EOL tests to ensure a full and error-free operation
9. Activation and deactivation like transport mode. (The definition of the transport mode refer to Functional Technical Specification Transport Mode Function)

The description can for example consist of 'reading of measurement values', 'reading of



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general data', 'writing of data', 'invoking of InputOutputControls-Routines' and so on.

If some function which is only designed for production purposes, effective locking mechanism must be have to ensure this function would not be implemented after production.

Any calibration setting that is carried out during production or aftersales using UDS services i.e. 31, 2E, etc. shall be done behind security unless these are part of the user defined calibration/settings.



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3.22 Aftersales Process

Following the sale of a vehicle if there is module replacement required or it has to be recalibrated/configured, the following process defines the method by which it can be achieved this is to ensure the ECU behaviour is consistent as intended by Vinfast/ECU supplier.

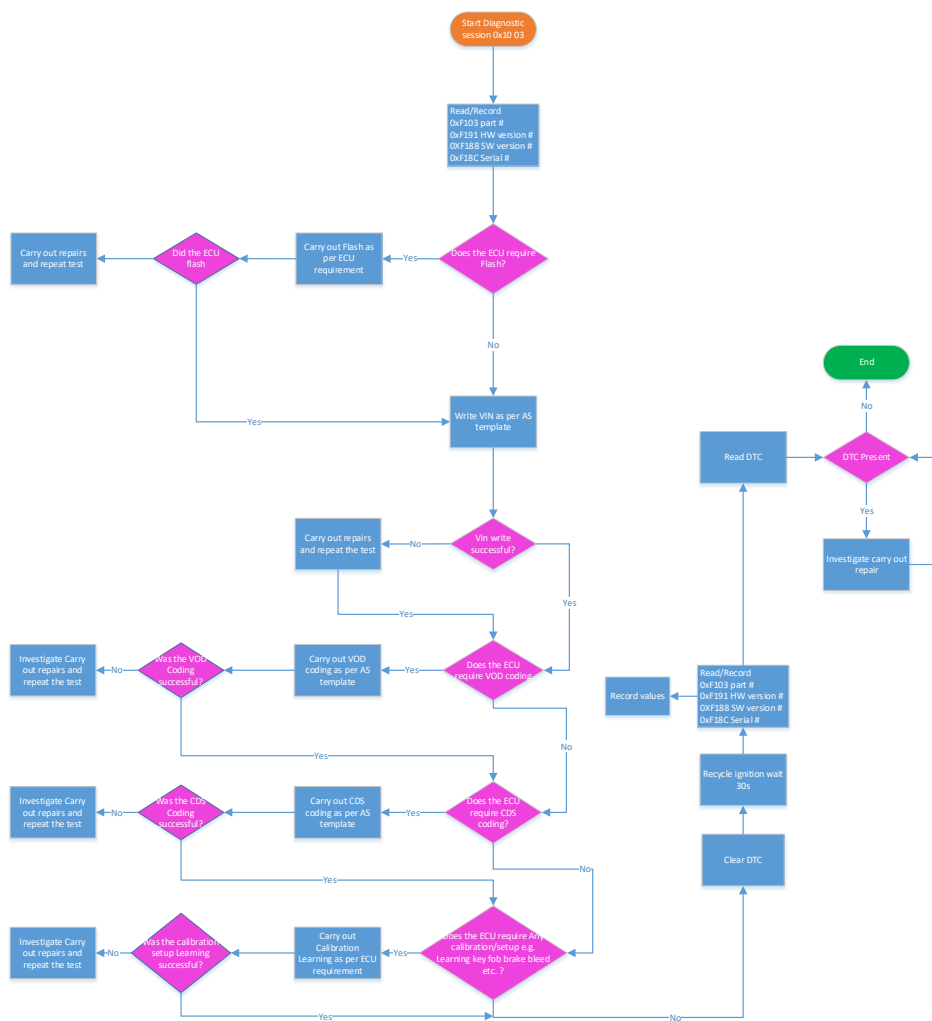


Fig. 4



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3.23 End of Line test sequence

The following process shall serve as a EOL sequence that all ECUs which are fitted to the vehicle will need to satisfy. Any calibration or coding sequence which the ECU needs in order to perform its intended function the supplier will need to detail this in the relevant ODX/EOL/AS documents describing how it should be carried out.

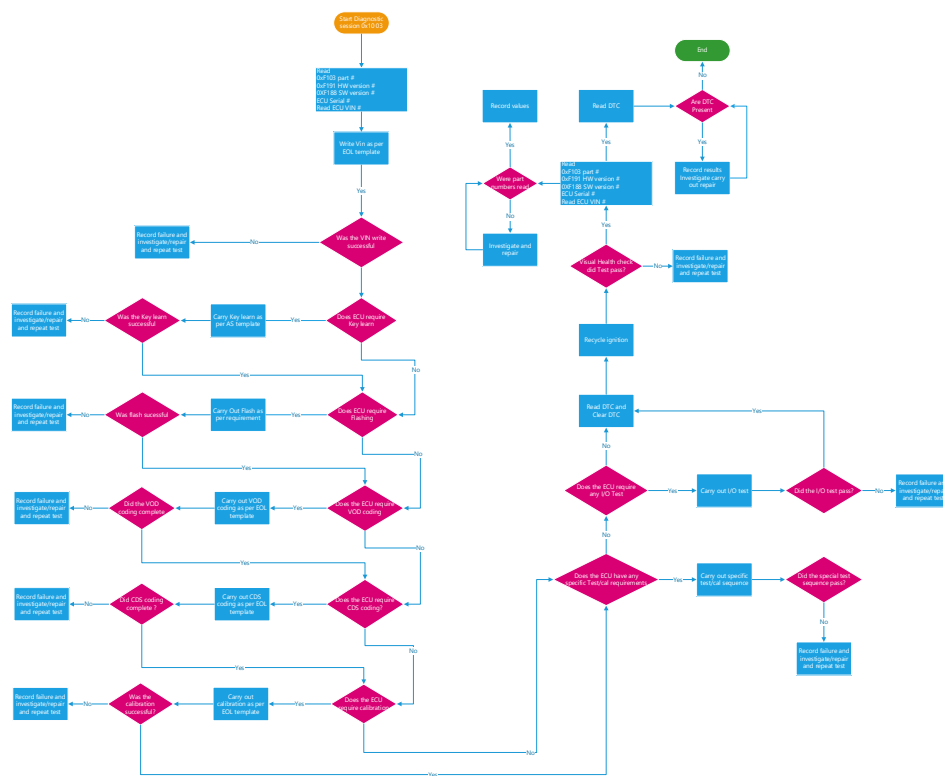


Fig. 5

3.24 LIN Diagnostic

In case of connected LIN ECUs, the Master and Slave ECUs should provide the LIN communication regarding ISO 17987. Dependent of the functionality of the slaves, the supplier of the relevant ECUs has to support LIN diagnostic category I, II or III. Detailed information refers to the LIN specification Ref [24].



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3.25 Diagnostic system and strategy

Overall Diagnostic system and strategy for vehicle faults including capturing, retrieving, and communicating these faults shall support the agreed VinFast customer and connected service models for all markets including any OTA & Remote Diagnostic functions.

Changes in adherence to any diagnostic standards or changes in standards that may lead to a change in compatibility with current or planned diagnostic tool chains must be agreed with Aftersales Engineering prior to the implementation of these standards.

3.26 DTC Selection & Usage (Fault identification and naming)

Think of the service technician who will be using this information to resolve faults. DTC numbers and names need to reflect the serviceability of the system being controlled and the systems it interacts with.

For example

- DTC numbering & naming shall reflect industry or VinFast standard definitions where available to allow for
 - Easy identification of common faults between ECU's.
 - Familiarity/consistency across VinFast products.
- Failure-type (Sub-type) shall best reflect the type of fault being diagnosed to allow system faults to be easily identified.
- Any unexpected behaviour caused by any ECU faults shall be diagnosed.
- Communication with other ECU's and any communication system (i.e. CAN-BUS) shall be diagnosed.
- All serviceable components of a system that is sensed or controlled by an ECU shall be diagnosed.
- Embedded components within an ECU or system that are not serviceable do not need to be individually diagnosed but should be covered by single or multiple system level DTC's for the ECU or system.
- If the ECU itself is embedded in a higher-level system/component, then the diagnosis of faults should reflect higher level system DTC's rather than more comprehensive individual sensed or controlled DTC's that would all lead to the same repair action i.e., replacement of the higher-level system/component.

3.27 Service Repair Actions

- It should be assumed that technicians will investigate issues relating with power source/fuses/grounds as part of a general check prior to conducting any specific DTC repair action.
- The repair action shall consider the smallest component that is replaceable as part of any repair action that has been agreed with Aftersales serviceability of the system/component.
- The repair instruction shall only refer to replacing serviceable components, which may be different to production and engineering environments. For example, we shall not instruct the technician to open an ECU and perform electronic repairs/analysis if the internal components are not serviceable and not released as replaceable item by the Aftersales service team. The component owner or Aftersales team should be able to best advice on this topic.



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- The repair action shall get to the point where a repair or component/assembly change is required to the vehicle. For example, cycling the ignition is not a repair action. Think about what has to happen next if this doesn't work? There can be multiple paths/workflows for a given DTC so all options shall be included.

3.28 Input & Output (incl. dynamic DID's) Controls

All inputs and outputs (physical and logical) at the serviceable system level shall be readable, outputs shall also be controllable where relevant.

3.29 Service Routines

The needs of each ECU and its serviceable system level will determine which routines are required to satisfy its service functions (i.e. replacement, re-programming, configuration, calibration, initialization, learning, self-diagnosis, etc.).

3.30 ECU Traceability and VIN Storage

The ECU requirement of Traceability **shall** be handled through VIN a 17-character unique identification number made of letters and numbers that uniquely identifies vehicle and this **shall** be implemented and stored consistently and correctly within all relevant ECUs such that common EOL and service procedures. This shall be applicable on all ECUs that can support a full or partial diagnostic server functionality and meet one or more of the use cases below: -

- 1 Have a requirement to store VIN to meet requirements of the insurance industry.
- 2 Have a requirement to detect and / or protect against an ECU being moved from one vehicle to another – This will normally be identified during the ECU FMEA process.
- 3 Have a requirement to report the vehicle VIN to meet emissions legislation of various market in which the vehicle is likely to be sold/operated.
- 4 Communicate using I/P, (Ethernet, Broad-R-reach, or some other method of communication between ECUs that is capable of exchanging diagnostic information etc.), and or takes part in the DOIP vehicle discovery process.

VIN Storage Use Cases:

Case 1: Storing VIN to meet the requirements of the insurance industry. An ECU will need to meet this use case if a requirement to permanently store VIN once has been cascaded from Vinfast through to EOL tool connected to the vehicle and thus configuring each applicable ECU to write the VIN using diagnostic commands. The VIN storage requirement is driven by the need to reduce the theft insurance rating of the vehicle by making original VIN traceable by the criminal investigation authorities using diagnostic services and thus allowing stolen and disguised or broken up vehicles to be identified. This is sometimes referred to as the "THATCHAM" requirement. There is no independent diagnostic requirement to store VIN in this way but where the security requirement exists, then there is a diagnostic requirement for the way this VIN storage is implemented.

A VIN learned according to Case 1 shall be readable with DID 0xF190 using the format defined in the Reference document, (Basic DID requirement). Prior to storing a VIN, F190 shall read all 0x00 (ASCII Null character). Note: DID F190 reports 24 characters using ASCII



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format and the VIN only uses the first 17 of these characters. In accordance with the requirements for data type ASCII. A VIN learned according to Case1 shall not be used for any other purpose by the ECU. The ECU shall remain fully functional regardless of whatever value of VIN is stored according to Case1, whether it matches the vehicle VIN or not, or even in the case if a VIN has not been stored at all. No DTCs shall be recorded/logged for a VIN mismatch since there will be no customer observable symptom associated with misaligned VIN.

Case 2: Storing VIN as a method of detecting if an ECU has been moved between vehicles. An ECU will need to meet this use case where an FMEA or risk analysis has shown that moving an ECU that had been set up to operate on a specific vehicle to another vehicle might result in undesirable ECU behaviour. For example, if a steering angle sensor had been calibrated to the straight ahead position in one vehicle and if then the sensor ECU was moved to another vehicle the calibration would become invalid and therefore, ECUs using the sensor data may then function incorrectly. A method to prevent this is to store the VIN in a (DID 0x0803) at the point an ECU is set up and then confirm the vehicle VIN is unchanged each time the ECU operation cycle is started this will only apply. Where there is a requirement to identify if an ECU has been calibrated or locked to a specific vehicle and then been moved to another vehicle then the ECU calibration or lock process shall be designed to store all or part of the vehicle VIN in the non-volatile memory as part of that process. (currently the whole VIN is not transmitted by the BCM hence only the last characters of the VIN will be compared to ensure that the ECU is still connected to the same vehicle). This stored VIN shall not be readable with DID F190. Each calibration / lock / build stage process that can be invoked separately shall store its own copy of the VIN and shall not be shared with any other ECU function that uses VIN including any storage under Case 1.

It maybe, the case where it's preferable to show the VIN that the ECU vehicle is locked to or where the calibration was last learned in which case the calibration VIN shall be stored in a different DID, under normal circumstances both DIDs shall mirror each other, therefore any ECU that needs to store linked VIN to calibration will need to subscribe to VIN networked ECU that have stored a VIN according to Case2 and which have a functional requirement to detect the ECU being moved between vehicles shall check this stored value against the transmitted VIN value as soon as possible after the start of each vehicle operation cycle once new VIN data has been correctly received. If a mismatch is detected the ECU shall take action according to its functional specification. If that action includes setting a warning visible to the vehicle driver, then a DTC shall be set that indicates the system is not calibrated or that an incorrect ECU is fitted. If the action is that recalibration is carried out automatically in the background, then if customer symptoms or warnings are present during this automatic process a suitable DTC shall be set but otherwise no DTC shall be set. The DTC shall not state that the VIN is incorrect only to indicate that the calibration is required of the affected ECU otherwise this will prompt an investigation which is not needed.

ECUs that have stored a VIN according to Case 4 shall determine a policy of whether the ECU is allowed to be updated / recalibrated to align with the new vehicle or whether once locked to a vehicle the ECU is never allowed to operate in another vehicle. In either case suitable control routines shall be implemented to store the VIN either once when the ECU is first fitted / calibrated or every time the calibration is re-run as part of the calibration process or as part of a control routine that exits a plant build mode or other similar special operation mode. ECUs shall not be implemented to store the VIN automatically (e.g. on first ECU power up) except where an associated re-calibration is also triggered automatically.



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Note: where the ECU is locked permanently to a vehicle, consideration should be given to the cost of scrap ECUs when recovering from misguides where vehicles are initially loaded electronically with a VIN that does not match the physical VIN plates.

Case 3: - Storing VIN so it can be reported to OBD scan tools as required to meet emission legislation.

An ECU will need to meet this use case where the ECU has a separate requirement to meet the SAE J1979 specification. Only a small sub-set of ECUs will have this requirement. ECUs with this requirement need to store the vehicle VIN that is (periodically transmitted over the vehicle network) so that this can be reported immediately to a scan tool without the need to wait for this periodic transmission.

This requirement is applicable only to emission relevant ECUs that are required to support legislated OBD scan tool communication according to SAE J1979 and are required to support OBD mode/service 0x09 info-type 0x02 (VIN). The Case is not applicable otherwise, ECUs that meet the criteria above shall store all 17 characters of the vehicle VIN as part of the vehicle configuration. Typically, ECU which fall in this category this will require storage of any changed data (similar to case 2) at power down and first use of the updated data in the next operation cycle. This check/update shall occur automatically on power down/shut down of the ECU. Apart from reporting this VIN using OBD mode/service 0x09 info-type 0x02 the same VIN shall also be readable using UDS (DID 0xF800) as defined in reference (12)

Case 4: - Storing VIN so it can be reported to a diagnostic tool during the DOIP vehicle discovery process. An ECU will need to meet this use case where the ECU supports diagnostic communication over internet protocol (DOIP) and has been designated as the DOIP edge node or a DOIP entity that takes part in the DOIP vehicle discovery process according to Reference Ref. (ISO 13400). ECUs with this requirement need to store the vehicle VIN that is periodically transmitted over the vehicle network (TBD) so that it can be reported immediately to a tester as part of vehicle discovery without the need to wait for this periodic transmission.

An ECU that has been designated as a DOIP edge node or a DOIP entity such that it takes part in the DOIP vehicle discovery process shall store all 17 characters of the vehicle VIN. Typically, this will require storage of any changed data at power down and first use of the updated data in the next operation cycle. This check / update shall occur automatically on power down/shut down of the ECU. A VIN stored to meet this requirement shall use a storage area separate from any VIN storage to meet Case 1 or a ECUs that support a bootloader, the VIN storage area shall be accessible by the ECU bootloader, as specified by the implementer is permitted to share the VIN stored to meet this Case with storage requirements implemented to meet use case 4 so long as no conflict between the use cases has been identified. Where the VIN2 storage is shared in this way the stored VIN will be readable using (DID 0xF801) as required by SAEJ1979. In cases where the VIN storage cannot be shared and optionally in all other cases the VIN4 shall be readable using (DID 0xF802).

Note: Case 4 This requirement is only expected apply to ECUs that are connected to a network that can support TCP or UDP Ethernet protocols.



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3.31 Programming in Service

For any ECU that is re-programmable in service either for ECU replacement or for software updates, all data files necessary for programming must be released in EBOM (i.e. flash driver, software and calibration files). Even if these files are not used by ENG or MFG. Part number and version control of these files must follow VinFast spec books and guidance documents, particularly around changes from saleable builds and beyond. Identification and storage of these files must follow agreed methods for Aftersales service and OTA.

3.32 ECU Diagnostic Information Templates

ECU diagnostic & programming information shall be provided in a format approved by Vinfast. Must be in English language only. Must include the following DTC information layout:

Property	Type	Description	Usage
DTC Number (hex)	3 byte	Per ISO15031	M
DTC Number	String	Per ISO15031 conversion for 3-byte format (eg 0xC14087 = U0140-87)	M
Failure Type Byte	1 byte	Per ISO15031/SAE-J2012DA	M
DTC Name	String	Description of the DTC + Failure Type Byte Description	M
Priority	Enumeration	Defines the priority of a DTC. 0 - highest priority 1 - medium priority 2 - low Reference DTC severity- ISO14229	M
Lamp Flag	Boolean	Indicates if a lamp indicator is requested (or driver information warning message is displayed)	M
Repair action	String	A textual description/workflow of how to solve the problem, related to the DTC.	M
Monitor enable conditions	String	A textual description of the conditions, required to execute the DTC's monitor correctly.	M
Monitor type	Enumeration	Defines whether the DTC's monitor is executed either continuously or by an event.	M
Monitor rate	String	A textual description on how often a DTC is checked (continuously), or when the DTC is checked (based on an event).	M
Test Failed criteria	String	A textual description of the conditions for the DTC monitor to produce test result "Test failed".	M
Mature Time	String	How long fault is present before DTC is triggered	M
Test Pass criteria	String	A textual description of the conditions for the DTC monitor to produce the test result "Test passed".	M
Service relevant	Enumeration	De-fines whether a DTC is relevant for service (Yes/No)	M
Demature Time	String	When all test pass criteria are met again, how long does it take for a Current DTC to transition to History status	M
Limp-home action	String	A textual description (required for service) of the remedial action applied to the performance of the ECU, component or vehicle.	U
DTC Aging criteria	String	A textual description of the conditions or cycles required to appraise the DTC as aged	M



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Local snapshot record	String	The information should be stored when the DTC occurrence	U
Vehicle Safety Level	String	0 - Normal 1 - Slight Fault 2 - Minor Fault 3 - General Fault 4 - Serious Fault 5 - Fatal Fault	U

Table 15: Layout of DTCs

All additional amounts for the DTC implementation, will be defined by Diagnostic department, based on ISO/SAE conformity.

4 Diagnostic Communication

4.1 Communication Speed

The Bus speed of Diagnostic-CAN is 500kbps.

4.2 Can Network Layer Parameter

For CAN Network Parameters please see CAN specification requirement (VFDSXVEEP0047).

4.3 Diagnostic Layer Parameters

Diagnostic layer Parameter	Min	Max	Timeout	Boot	Unit
P2server	0	50	n/a	25	ms
P2client	n/a	n/a	150	150	ms
P2*server	0	2000	n/a	2000	ms
P2*client	n/a	n/a	5000	5000	ms
S3 server	n/a	n/a	5000	5000	ms
S3client	0	3000	4000	4000	ms
P4 CAN Server	0	5000	5000	5000	ms
Network layer Parameter	Timeout Value		Performance requirement		
N_As/N_Ar	70ms		--		
N_Bs	150ms		--		
N_Br	--		$(N_{Br} + N_{Ar}) < 70ms$		
N-Cs	--		$(N_{Cs} + N_{As}) < 70ms$		
N_Cr	150ms		--		
ECU Flow Control Parameters	Application Mode		Boot Mode		
BS	8		0		
STmin	20ms		0ms direct and up to 5ms max on the Vehicle going through XGW depending on the min process time of the XGW.		

Table 16: Requirement of Diagnostic and Network Layer Parameter (note 5)

\$ Note: timing requirement for the above table taken from (ISO-15765-3 ISO-15765-2) and shall always be referred for implementation where required.

S3 Server time: -

Keep a diagnostic session other than default session active whilst not receiving any diagnostic messages for a maximum of 5000ms time out. S3Server timer handling is based on the network layer service primitives, which means that the S3Server timer is also restarted upon the reception of a diagnostic request



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message even if that message is not supported by the server.

S3 Client: -

Time between functionally addressed Tester Present (3E hex) request messages transmitted by the client to keep a diagnostic session other than the default Session active in multiple servers (functional communication) or maximum time between physically transmitted request messages to a single server Physical communication). Min 0ms Max 3000ms Time out 4000ms.

S3 Timer: -

Where there is communication between ECU and tester as long as the time between communication does not exceed the P2*_server_max time of 2000ms it should maintain session and reset the S3_timer after each communication that is able to reach the network layer, this should be carried out even if the service is not supported by the ECU. Please refer to ISO15765-3 for clarification.

Any time a server that is in the process of handling any diagnostic service, it stops its S3 timer. When the diagnostic service is completely processed then the server restarts its S3 timer.

P2 Server

Performance requirement for the server to start with the response message after the transmission of a negative response message with response code 78 hex (enhanced response timing). The ECU must respond with either a positive or negative response within a maximum 5000ms even if the message was sent with suppress positive response bit set.

Time between tester request message and the receipt of an un-segmented response message or first frame of a segmented response message

The ECU shall start sending its response message within P2CAN after the Request message has been correctly received. P2CAN_Server 50ms.

4.4 Diagnostic services response times

An ECU shall always execute and complete a request before transmitting a response unless specified otherwise. If the time required to complete a request is longer than the specified value P2CAN_Server, then the ECU may respond with negative response code 0x78 "request Correctly Received Response Pending" as a method of delaying its final response. The timing parameter P2CAN_Server is specified in table 16. Note that the maximum value of P2CAN_Server is fixed and it's the same in the default Session and extended Diagnostic Session.

If a diagnostic Session Control (0x10) or ECU Reset (0x11) request requires a reset of the ECU, the positive response shall be made before carrying out the requested action. Until the requested action is complete the ECU shall either ignore any received diagnostic requests or respond with a negative response code of Busy Repeat Request (0x21). Note: see ISO14229-2 2013 which specifies the maximum time during which the ECU is permitted to be unresponsive to -request messages following a reset of the ECU. Generally, a CAN ECU should be ready within 500ms after it has given a positive response to reset request.

If a diagnostic Session Control service 10) or ECU Reset (0x11) request does not require a reset of the ECU. The ECU shall complete the functionality simultaneously with or immediately following the positive response so that the requested functionality is guaranteed to be complete prior to the arrival of the next possible diagnostic request by the application.)

If a request is sent with the Suppress Positive Response Message (SPRM) Indication Bit set to true, the ECU shall execute and complete the request within P4CAN server time (5000ms). Note



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that when negative response code 0x78 is utilised for a given request, the ECU shall always send its final response to that request (i.e., positive response or negative response message with negative response code other than 0x78), which is independent from the value of the PRM Indication Bit.

4.5 Diagnostic Layer Error Handling

Error handling for the application layer and session management to be fulfilled by the client and server during physical and functional communication shall be in accordance with the ISO 15765- 3.

The client error handling defined shall be performed for a maximum of two times, which means that the worse-case of service request transmissions is three.

Additionally, if there is a server timeout, the server must return to default session immediately. When an ECU is sent a request that it is processing it should not reply back with response back unless it is ready to process the next request, exception to this requirement is when the request is for a ECU reset in which case the ECU is required to send the response and then carry out the reset. Whilst recovering from reset ECU shall not respond to any further requests unless it is ready to process the request and it should not buffer any requests. This is to prevent the ECU from buffering requests as multiple request from the client may result in the ECU being overwhelmed and may cause it to lock up. The maximum period for a CAN ECU to be non-responsive following a reset is 500ms.

4.6 Functional Addressing

Functional addressing Servers All the requests with length less than eight can be physically or functionally addressed

- All the requests with length greater than seven must be physically addressed.
- Where Servers do not support the request replace the negative responses containing NRCs \$11, \$12, or \$31 with "No Response" for all functionally addressed requests.
- Where the Sever response is greater than 8bytes then the response shall be given in a segmented response.

4.7 Data length code (DLC)

The CAN DLC contained in every diagnostic CAN frame shall always be set to eight (8). The unused data bytes of a CAN frame shall be padded with zero (0). Any diagnostic CAN frame with a DLC less than eight (8) shall be ignored.

Any functional request CAN frame with dynamic DLC can be less than defined DLC of fixed length. Where a packet is missing will lead to ECU declaring missing message error.

4.8 Addressing method

For the current Project follow addressing method is required:

- Normal Addressing

For this method the requirements of the ISO 15765 are binding. The using of the method is described in the ISO 14229 for UDS ECUs. Deviations must be coordinated with the Project Architecture and Diagnostic Team.

4.9 ECU Address

Each ECU in CAN network has one specific address. This is necessary to send the service request



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to the correct ECU. For special services, a functional address (0x6FF) will be used. This broadcast address is needed for requests to all ECU's (e.g. UDS Service \$19 use a functional address, to reading DTC's from all ECU's in the network with sending the request only one time). The addresses of all ECU's in the network have to be defined by the network manager.

In case of different components and communication protocols on the CAN Bus systems the addressing method and calculation of Response IDs are different. For further information, refer to the document below:

- ECU_Identifier_Specification

For OBD CAN Identifiers the requirements of the ISO 15765 (part 4) are binding.

Deviations shall agree by the Project department.

5 System Description

5.1 General System Description

The BUS Architecture include different BUS systems. For further information, refer the document:

System_schematics for the vehicle.

5.2 CAN Network

For the CAN network the requirements of the CAN Network Specification is binding:

- CAN_Specification [ref 6]

5.3 LIN Network

For the LIN network the requirements of the LIN Specification is binding
For further information refer the document:

- LIN Specification ISO 9141-2 for guidance

5.4 Specific Requirements

All ECUs in the Project have to store internally information about Hardware, Supplier, Software, ECU serial number etc. This are common Data Identifier. For further information, refer the document:

- Basic Services DID PID Definition



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6 General Diagnostic Specification

6.1 Diagnostic Protocol

The communication between car and tester must comply with the requirements of Unified Diagnostic Services UDS ISO 14229-1-2020.

A special part of the ON-Board diagnosis is the fault detection of emissions-related parts regarding OBDII/EOBD. Here are the special requirements of the ISO 15031 (part 1 ... part 7) are binding.

6.2 Diagnostic Data

The defined data exchange format for all ECUs is an ODX package. This data package has to include all necessary diagnostic information of the ECUs. The Supplier has to deliver a fully tested file regarding ODX standard ISO 22901 as Version 2.2. The Supplier makes sure, that the file is running and can be used on a valid MCD Server regarding ISO 22900.

For special tasks like Flash and Coding support, the supplier delivers a fully functional ODX data file.

6.3 Diagnostic Sessions

For diagnostic task, the ECU must support different sessions. This chapter introduces the different diagnostic sessions, which should be available for the relevant ECUs.

Requirement about the common Service and Subservice are described in the ISO14229 and CES Basic Services DID PID Definition.

6.4 Default Session

The Default Session shall be enabled in the ECU, if no other session is requested after power up.

6.5 Extended Session

The Extended Session shall be enabled in the ECU, if this diagnostic tasks requested by the tester. This session shall be maintained for the duration of the P2*server_max time out of 2000ms unless a subsequent request is sent which will reset the timer before P2* Server_max timeout occurs.

6.6 Coding Session

The Coding Session shall be enabled, if the Coding process requested by the tester. Please refer to the coding specification for the specific details.

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6.7 Programming Session

The Programming Session shall be enabled, if the Flash/Programming sequence is requested by the tester, please refer to flash specification for specific details.

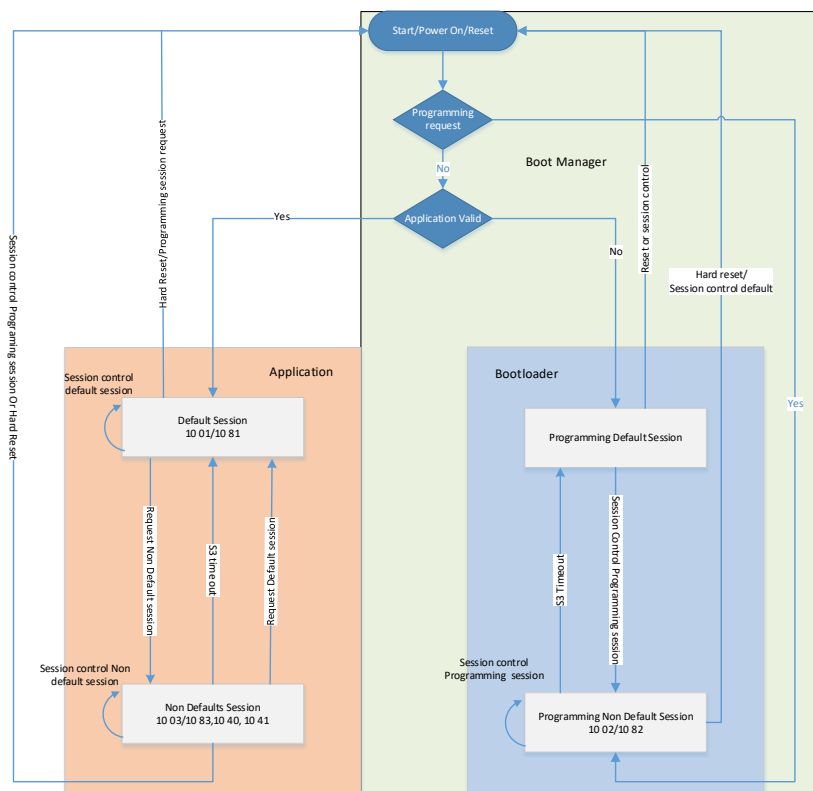


Figure 6: Session Control



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6.8 Security Level for ECU Access

To protect against unauthorized access to the vehicle electronic system, the service Security Access (\$27) method shall be used.

For each of the Security Levels defined in table 12 and table 13 below, the Security Access (\$27) Keys will be unique

Security Level	Description	Key must defined by OEM	Usage
0	No Level needed or OBD relevant amounts	no Key needed	M
1	After Sales (Service)	RND and OEM Key	M
2	Manufacturing	RND and OEM Key	M
3	ECU-Coding	RND and OEM Key	O
4	ECU-Programming (Flash)	RND and OEM Key	O

Table 17: Security Level

Security Level	Description	Request Seed	SendKey	Session	usage
0	No Level needed or OBD relevant amounts			Default	M
1	After Sales (Service)	0x01	0x02	Extended	M
2	Manufacturing	0x03	0x04	Extended	M
3	ECU-Coding	0x05	0x06	Coding	O
4	ECU-Programming (Flash)	0x07	0x08	Programming	O

Table 18: Seed and Key Subservices

For full Security Access (\$27) specific requirements, please refer to VF_Security_Access_Algorithm specification.

6.9 Diagnostic Services

The diagnostic service is the summary of a request message and a response message. Any diagnostic service has an explicit service identifier.

- Type request
- Type response
- Type positive response
- Type negative response

Each service request by a Tester must be answered by the ECU with a valid response. It is not allowed, that the ECU does not answer to the Tester request.

6.10 Diagnostic Service Example Simple Communication

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Simple Communication in UDS without Sub- Function Byte

Example

- 1 Positive Response
- 2 Negative Response

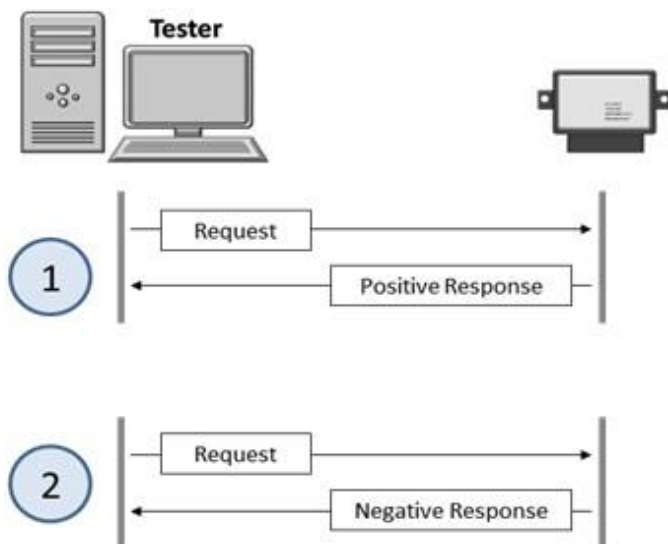


Figure 7: Simple Communication



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6.11 Diagnostic Services UDS

This chapter describes the Diagnostic services for the UDS protocol.

All mandatory services in this chapter shall be supported by the ECU supplier. The ECU Shall support suppress positive response requests in all services where this is deemed helpful; this is particularly useful where the use of functional addressing is required to carry out the same task across the whole vehicle with just a single command i.e. to change the session of the whole vehicle or to clear DTCs etc. An overview about the requested services is in the table 14 below:

SID hex	Service Name	Usage
\$10	DiagnosticSessionControl	M
\$11	ECUReset	M
\$27	SecurityAccess	M
\$28	CommunicationControl	M
\$3E	TesterPresent	M
\$83	AccessTimingParameter	U
\$84	SecuredDataTransmission	U
\$85	ControlDTCSetting	M
\$86	ResponseOnEvent	U
\$87	LinkControl	U
\$22	ReadDataByIdentifier	M
\$23	ReadMemoryByAddress (will not be allowed due to security concerns)	NA
\$24	ReadScalingDataByIdentifier	U
\$2A	ReadDataByPeriodicIdentifier	U
\$2C	DynamicallyDefineDataIdentifier	U
\$2E	WriteDataByIdentifier	M
\$3D	WriteMemoryByAddress (will not be allowed due to security concerns)	NA
\$14	ClearDiagnosticInformation	M
\$19	ReadDTCInformation	M
\$2F	InputOutputControlByIdentifier	U
\$31	RoutineControl	M
\$34	RequestDownload	U
\$35	RequestUpload (will not be allowed due to security concerns)	NA
\$36	TransferData	U
\$37	RequestTransferExit	U
\$29	Authentication	M
\$38	File Transfer where ECU supports file transfer	M

Table 19: Table of Services ISO 14229 UDS



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6.12 \$10 DiagnosticSessionControl

The client requests to control a diagnostic session with a server(s).

Diagnostic Session Control Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	Diagnostic Session Control Request Service Id	M	10
Sub Function = []			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [diagnosticSessionType]	M	00 - FF
	defaultSession	M	01 /81
	programmingSession	M(CAN node)	02 /82
	extendedDiagnosticSession	M	03 /83
	codingSession	M	41/C1
	Bench session	U	40/C0
	Roller Session	U	50/D0

Table 20 Diagnostic Session Control

6.13 \$11 ECU Reset

The client forces the server(s) to perform a reset, the server will send a response back and then carry out the reset as requested, whilst undergoing the reset the server shall be not responsive for a duration no longer than 500ms.

ECUReset Request Service Id \$11			
Data Byte No	Parameter Name	Usage	Data Value
1	ECUReset Request Service Id	M	11
subFunction = []			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [resetType]	M	00 - FF



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	hardReset	M (programming)	01/81
	keyOffOnReset	C	02/82
03	softReset	M	03 /83

Table 21 ECU reset

6.14 \$27 SecurityAccess

The client requests to unlock a secured server(s).

Security Access Request Service Id \$27			
Data Byte No	Parameter Name	Usage	Data Value
1	SecurityAccess Request Service Id	M	27
subFunction = [securityAccessType = requestSeed]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [securityAccessType = requestSeed]	M	00 – FF
	requestSeed (afterSale)	M	01
	requestSeed (EOL)	M	03
	requestSeed (Coding)	M	05
	requestSeed (flashing)	M	07
subFunction = [securityAccessType = sendKey]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [securityAccessType = sendKey]	M	00 – FF
	sendKey	M	02
	sendKey	M	04
	sendKey	M	06
	sendKey	M	08
3	securityKey[] = [key#1 (high byte)	M	00 - FF
:	:	:	:
#n	key#m (low byte)]	U	00 - FF



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Table 22 Security Access



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6.15 \$28 CommunicationControl

The client requests the server to control its communication.

CommunicationControl Request Service Id \$28			
Data Byte No	Parameter Name	Usage	Data Value
1	CommunicationControl Request Service Id	M	28
subFunction = [controlType]			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [controlType]	M	00 - FF
	enableRxAndTx	M	00 /80
	enableRxAndDisableTx	M	01 /81
	disableRxAndEnableTx	U	02 /82
	disableRxAndTx	M	03 /83
communicationType			
Data Byte No	Parameter Name	Usage	Data Value
3	communicationType	M	00 - FF
	normalCommunicationMessages	M	01
	networkManagementCommunicationMessages	M	02 *
	NormalandNetworkManagementCommunicationM essages	M	03*
* 1, If an ECU implements network management function, 02 shall support this communication type for communicationControl service. 2, For flashing function, 03 should be supported by all the ECUs.			

Table 23 Communication control



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6.16 \$3E TesterPresent

The client indicates to the server(s) that it is still present.

TesterPresent Request Service Id \$3E			
Data Byte No	Parameter Name	Usage	Data Value
1	TesterPresent Request Service Id	M	3E
subFunction = [zeroSubFunction]			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [zeroSubFunction]	M	00 - 7F
	zeroSubFunction	M	00 /80

Table 24: Tester Present

6.17 \$83 Access Timing Parameter

The client uses this service to read/modify the timing parameters for an active communication.

AccessTimingParameter Request Service Id \$83			
Data Byte No	Parameter Name	Usage	Data Value
1	AccessTimingParameter Request Service Id	U	83
subFunction = [timingParameterAccessType]			
Data Byte No	Parameter Name	Usage	Data Value
2	sub-function = [timingParameterAccessType]	M	00 - 7F
	readExtendedTimingParameterSet		01
	setTimingParametersToDefaultValues		02
	readCurrentlyActiveParameters		03
	setTimingParametersToGivenValues		04
timingParameterRequestRecord			
Data Byte No	Parameter Name	Usage	Data Value
3	TimingParameterRequestRecord[Byte#1	C	00-FF
	:	:	:
#n	Byte#m]	C	00-FF

Table 24 Access Timing



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\$85 Control DTC Setting

The client controls the setting of DTCs in the server.

ControlDTCSetting Request Service Id \$85			
Data Byte No	Parameter Name	Usage	Data Value
1	ControlDTCSetting Request Service Id	M	85
subFunction = []			
Data Byte No	Parameter Name	Usage	Data Value
2	subFunction = [DTCSettingType]	M	00 - FF
	on	M	01 /81
	off	M	02 /82

Table 25: Control DTC setting

6.18 \$22 ReadDataByIdentifier

The client requests to read the current value of a record identified by a provided data identifier

ReadDataByIdentifier Request Service Id \$22			
Data Byte No	Parameter Name	Usage	Data Value
1	Read Data By Identifier Request Service Id	M	22
dataIdentifier[] #1 = []			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier[] #1 = [byte#1 (MSB)	M	00 - FF
3	byte#2]	M	00 - FF
:	:	:	:
n-1	dataIdentifier[] #m = [byte#1 (MSB)	U	00 - FF
n	byte#2]	U	00 - FF

Table 26: Read Data By ID



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6.19 \$23 ReadMemoryByAddress

The client requests to read the current value of the provided memory range.

ReadMemoryByAddress Request Service Id			
Data Byte No	Parameter Name	Usage	Data Value
1	Read memory By address Request Service Id	U	23
AddressAndLengthFormatIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
2	AddressAndLengthFormatIdentifier	M	00 - FF
MemoryAddress			
Data Byte No	Parameter Name	Usage	Data Value
3	MemoryAddress[] = [byte#1 (MSB)	M	00 - FF
:	:	:	:
6	byte#4]	M	00 - FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
7	MemorySize[] = [byte#1 (MSB)	M	00 - FF
8	Byte#2	M	00 - FF

Table 27: Read Memory by Address



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\$2A ReadDataByPeriodicIdentifier

The client requests to schedule data in the server for periodic transmission.

ReadDataByPeriodicIdentifier Request Service Id \$2A			
Data Byte No	Parameter Name	Usage	Data Value
1	Read Data By Periodic Identifier Request Service Id	U	2A
TransmissionMode[]			
Data Byte No	Parameter Name	Usage	Data Value
2	transmissionMode[]=[byte#1]	U	00 - FF
	sendAtSlowRate	U	01
	sendAtMediumRate	U	02
	sendAtFastRate	U	03
	stopSending	U	04
PeriodicDataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value
3	PeriodicDataIdentifier[]#1	C	00-FF
:	:	:	:
#m+2	PeriodicDataIdentifier[]#m	U	00-FF

Table 28 Read data by periodic identifier



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6.20 \$2C DynamicallyDefineDataIdentifier

The client requests to dynamically define data Identifiers that may subsequently be read by the readDataByIdentifier service.

DynamicallyDefinitionDataIdentifier Request Service Id \$2C			
Data Byte No	Parameter Name	Usage	Data Value
1	DynamicallyDefinitionDataIdentifier Request Service Id	U	2C
Definition type[]			
Data Byte No	Parameter Name	Usage	Data Value
2	definitionType[]=[defined by identifier]	U	01
dynamicallyDefinedDataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value
3	dynamicallyDefinedDataIdentifier []#1	U	00-FF
4	dynamicallyDefinedDataIdentifier []#2	U	00-FF
sourceDataIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
5	sourceDataIdentifier []#1	U	00-FF
6	sourceDataIdentifier []#2	U	00-FF
PositionInSourceDataRecord			
Data Byte No	Parameter Name	Usage	Data Value
7	PositionInSourceDataRecord	U	00-FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
8	memorySize [#1]	U	00-FF
:	:	:	:
sourceDataIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
n-3	sourceDataIdentifier []#1	U	00-FF
n-2	sourceDataIdentifier []#2	U	00-FF
PositionInSourceDataRecord			



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Data Byte No	Parameter Name	Usage	Data Value
n-1	PositionInSourceDataRecord	U	00-FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
n	memorySize [#1]	U	00-FF

Table 29: Dynamic defined Identifier

6.21 \$2E WriteDataByIdentifier

The client requests to write a record specified by a provided dataIdentifier.

WriteDataByIdentifier Request Service Id \$2E			
Data Byte No	Parameter Name	Usage	Data Value
1	WriteDataByIdentifier Request Service Id	M	2E
dataIdentifier[]			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier[] = [byte#1 (MSB)	M	00 - FF
3	byte#2]	M	00 - FF
dataRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4	dataRecord[] = [data#1	M	00 - FF
:	:	:	:
m+3	data#m]	U	00 - FF

Table 30: Write Data by ID



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6.22 \$3D WriteMemoryByAddress

The client requests to overwrite a provided memory range.

WriteMemoryByAddress Request Service Id \$3D			
Data Byte No	Parameter Name	Usage	Data Value
1	write memory By address Request Service Id	U	3D
AddressAndLengthFormatIdentifier			
Data Byte No	Parameter Name	Usage	Data Value
2	AddressAndLengthFormatIdentifier	M	00 - FF
MemoryAddress			
Data Byte No	Parameter Name	Usage	Data Value
3	MemoryAddress[] = [byte#1 (MSB)	M	00 - FF
:	:	:	:
6	byte#4]	M	00 - FF
MemorySize			
Data Byte No	Parameter Name	Usage	Data Value
7	MemorySize[] = [byte#1 (MSB)	M	00 - FF
8	Byte#2	M	00 - FF
dataRecord			
Data Byte No	Parameter Name	Usage	Data Value
9	dataRecord[]=[data byte#1	M	00 – FF
:	:	:	:
#m+9	byte#m	U	00 – FF

Table 31: Write Memory by Address



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6.23 \$14 ClearDiagnosticInformation.

Allows the client to clear diagnostic information from the server (including DTCs, captured data, etc.)

Clear Diagnostic Request Service Id \$14			
Data Byte No	Parameter Name	Usage	Data Value
1	ClearDiagnosticInformation Request Service Id	M	14
groupOfDTC[]			
Data Byte No	Parameter Name	Usage	Data Value
2	groupOfDTC[] = [groupOfDTCHighByte	M	FF
3	groupOfDTCMiddleByte	M	FF
4	groupOfDTCLowByte]	M	FF

Table 32: Clear DTC

6.24 \$19 ReadDTCInformation

Allows the client to request diagnostic information from the server (including DTCs, captured data, etc.)

Subfunctions of Service \$19:

Sub Service	Subservice Name	Usage
\$01	reportNumberOfDTCByStatusMask	M
\$02	reportDTCByStatusMask	M
\$03	reportDTCSnapshotIdentification	U
\$04	reportDTCSnapshotRecordByDTCNumber	M
\$05	reportDTCSnapshotRecordByRecordNumber	U
\$06	reportDTCExtendedDataRecordByDTCNumber	M
\$07	reportNumberOfDTCBySeverityMask	U
\$08	reportDTCBySeverityMaskRecord	M
\$09	reportSeverityInformationOfDTC	U
\$0A	reportSupportedDTC	M
\$0B	reportFirstTestFailedDTC	M
\$0C	reportFirstConfirmedDTC	U
\$0D	reportMostRecentTestFailedDTC	U
\$0E	reportMostRecentConfirmedDTC	U
\$0F	reportMirrorMemoryDTCByStatusMask	U
\$10	reportMirrorMemoryDTCExtendedDataRecordByDTCNumber	U
\$11	reportNumberOfMirrorMemoryDTCByStatusMask	U
\$12	reportNumberOfEmissionsRelatedOBDDTCByStatusMask	U
\$13	reportEmissionsRelatedOBDDTCByStatusMask	U



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\$14	reportDTCFaultDetectionCounter	U
\$15	reportDTCWithPermanentStatus	U

Table 33: Subfunction Service \$19

6.25 \$2F InputOutputControlByIdentifier

The client requests the control of an input/output specific to the server. For full implementation of enabling mask please see ISO14229-1(2020) table 400 section 13.2.2.3

InputOutputControlByIdentifier Request Service Id \$2F			
Data Byte No	Parameter Name	Usage	Data Value
1	InputOutputControlByIdentifier Request Service Id	M	2F
dataIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
2	dataIdentifier#1[] = [byte#1 (MSB)	M	00 - FF
3	byte#2 (LSB)]	M	00 - FF
controlOptionRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4	controlOptionRecord#1[] = [controlState#1/inputOutputControlParameter	M	00 - FF
:	:	:	:
4+(m-1)	controlState#m]	C	00 - FF
controlEnableMaskRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
4+m	controlEnableMaskRecord#1[] = [controlMask#1	C	00 - FF
:	:	:	:
4+m+(r-1)	controlMask#r]	C	00 - FF

For the inputOutputControlParameter, following parameters defined:

Hex	Description	Cvt
0x00	returnControlToECU	M
0x01	resetToDefault	U
0x02	freezeCurrentState	U
0x03	shortTermAdjustment	M



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Table: 34 I/O Control by ID

6.26 \$31 RoutineControl

The client requests to start, stop a routine in the server(s) or requests the routine results.

RoutineControl Request Service Id \$31			
Data Byte No	Parameter Name	Usage	Data Value
1	RoutineControl Request Service Id	M	31
sub-function = [routineControlType]			
Data Byte No	Parameter Name	Usage	Data Value
2	startRoutine	M	01
2	stopRoutine	U	02
2	requestRoutineResults	U	03
routineIdentifier []			
Data Byte No	Parameter Name	Usage	Data Value
3	routineIdentifier [] = [byte#1 (MSB)	M	00 - FF
4	byte#2]	M	00 - FF
routineControlOptionRecord[]			
Data Byte No	Parameter Name	Usage	Data Value
5	routineControlOptionRecord[] = [routineControlOption#1	C/U	00 - FF
:	:	:	:
n	routineControlOption#m]	C/U	00 - FF

Table: 35 Routine Control

6.27 \$34 RequestDownload

The client requests the negotiation of a data transfer from the client to the server.

6.28 \$35 RequestUpload

The client requests the negotiation of a data transfer from the server to the client. This service shall not be supported due security concerns

6.29 \$36 TransferData

The client transmits data to the server (download) or requests data from the server (upload).



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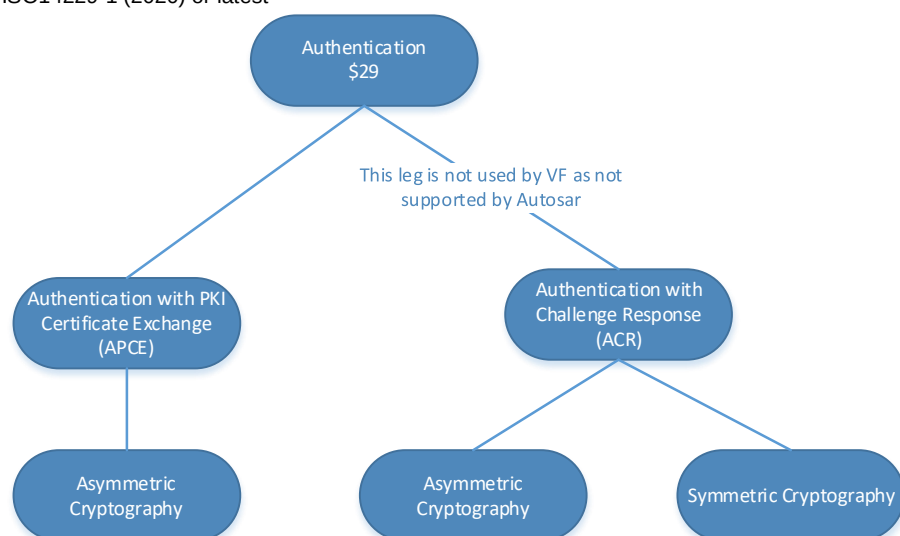
6.30 \$37 RequestTransferExit

The client requests the termination of a data transfer.

Note: - that all above services' definition must be compliant with ISO 14229 standard, including diagnostic request, positive response, all negative responses and negative response code(NRC) shall also be implemented in accordance with current definition of ISO requirements. Any deviation must be aligned with OEM diagnostic team prior to implementing these in software.

6.31 \$29 Authentication

For details on the description of how this service is fully implemented and used please refer to ISO14229-1 (2020) or latest





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Authentication \$29			
Data Byte No	Parameter Name	Usage	Data Value
1	Authentication Request SID	M	29
sub-function			
Data Byte No	Parameter Name	Usage	Data Value
2	DeAuthenticate (request to leave Authenticated state)	M	00
2	Verify Certificate Unidirectional (Initiate Authentication by verifying the Certificate).	M	01
2	Verify Certificate Bidirectional (Initiate Authentication by verifying the Certificate and generating a Proof of Ownership from the server).	M	02
2	Proof Of Ownership (Verify the Proof of Ownership from the client).	M	03
2	Transmit Certificate Verify Certificate and extract information from Certificate to handle it according to its contents.	M	04
2	Authentication Configuration Indicates the provided authentication configuration	M	08

6.32 DIDs

For processes in Production and/or in Service it is necessary to use a number of common basic DIDs, which will be used for identification of components, software or the complete vehicle. The number and content of this common DIDs must be clarified together with the Logistic Team, Production and Service and should have the same structure for all used ECUs. Deviations have to be agreed with OEM diagnostic team.

A minimum of common Basic DIDs are defined in the document:

- Basic_DID_ Definition

6.33 ECU Specific DIDs

All ECU specific Data/Parameter Identifier must be described in the respective component specification. And OEM ECU engineer and ECU supplier are responsible for the definition of DIDs' parameter, content, formula, format, unit and using method and details of its implementation shall be described in the ODX template to assist manufacturing Aftersales teams. Any DTCs which is triggered by specific event such as a measured voltage or a specific value of a signal shall support a DID indicating the measured value being reported.



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6.34 Supplier Diagnostic Quality Report

The Supplier of the ECU has to deliver an Diagnostic Quality Report. The content and design of the report is defined by the Diagnostic Department and should not be changed. The Supplier should update the Report for any Function Release Step and send the full filled report to the Integration team during the relevant Function Delivery step. The content of the Diagnostic Quality Report has to be filled completely by the Supplier. The Supplier has to fill the report truly and accurately.

7 Guidelines for ECU Diagnostics

7.1 Message to ECU Suppliers

- Maximize your ECU diagnostics to minimize the no fault found returns (warranty parts)
 - This applies to DTC's, DID's, RID's etc
- All I/O circuits shall be diagnosable against short to battery/ground/open circuit where possible. This will be dependent on ECU design
- All inputs (including Sensors) shall be supported in Service \$22
- All outputs shall be supported in Service \$2F or \$31 where possible
- ECU shall have a DTC for all unexpected vehicle behavior caused by
 - Plausibility/Rationality faults
 - Loss of Communications
 - Invalid Data Received from another ECU
 - Checksum and Alive Counter Errors detected in received CAN messages
 - Where ECU software determines it has not correctly performed one of its functions, or otherwise decides upon a vehicle reaction that a customer may notice

7.2 Assigning DTC numbers

- DTC's are assigned to an ECU per ISO-15031-6 / SAE-J2012DA, else for Manufacturer Controlled Ranges please follow VINFAST DTC Ranges spreadsheet (see below).
- Between VINFAST EE Diagnostic Group and ECU Supplier, aim to commonize DTC assignments across all ECU's.

Note reference Documents shall be checked for any updates or clarifications when using the content contained in sections 7.2 through to 7.4



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1 ECU DTC			DTC number range of DTC high&middle byte _ISO15031-6 (and J2012-DA) - ISO defined	DTC number range of DTC high&middle byte _ISO15031-6 Manufacturer controlled
System	System DTC	ECU		
Chassis	Cxxxx	EPS		C1300-C13FF / 0x5300-0x53FF
		ESC		C1500-C15FF / 0x5500-0x55FF
		VCU		P1000-P10FF / 0x1000-0x10FF
		BMS		P1200-P12FF / 0x1200-0x12FF
Powertrain	Pxxxx	EDS-F & EDS-R	Please refer to SAE J2012-DA	P1800-P18FF / 0x1800-0x18FF
		3in1(IPDM)		P1400-P14FF / 0x1400-0x14FF
		AVAS		P1600-P167F / 0x1600-0x167F
		EAC (BEV Plus)		P1680-P16FF / 0x1680-0x16FF
		GS		P1500-P157F / 0x1500-0x157F
		TCU		P1580-P15FF / 0x1580-0x15FF
		EMS		P1700-P17FF / 0x1700-0x17FF
				P1180-P11FF / 0x1180-0x11FF
				P1100-P117F / 0x1100-0x117F
		ACM		B1400-B14FF / 0x3400-0x34FF
		ADAS		B1900-B19FF / 0x3900-0x39FF
		APM		B1100-B117F / 0x3100-0x317F
Body	Bxxxx	BCM	Please refer to SAE J2012-DA	B1000-B10FF / 0x3000-0x30FF
		CCM		B1700-B17FF / 0x3700-0x37FF
		CPD		B1180-B11FF / 0x3180-0x31FF
		ETG		B1200-B127F / 0x3200-0x327F
		HUO		B1280-B12FF / 0x3280-0x32FF
		IPC		B1800-B18FF / 0x3800-0x38FF
		MFS		B1500-B15FF / 0x3500-0x35FF
		MHU		B1600-B16FF / 0x3600-0x36FF
		OCS		B1300-B137F / 0x3300-0x337F
		PSM-D		B1380-B13FF / 0x3380-0x33FF
		SHUJ-F		B2100-B217F / 0xA100-0xA17F
		SHUJ-R		B2180-B21FF / 0xA180-0xA1FF
Network	Uxxxx	ALL ECU		B2000-B20FF / 0xA000-0xA0FF
				U1100-U11FF / 0xD100-0xD1FF

Notes:
Step 1. DTC shall first be assigned according to the trouble code defined in ISO15031-6 (and J2012-DA).
Step 2. If the corresponding trouble code cannot be found, the code shall be selected within the scope defined by Manufacturer controlled range in column E.

Figure 8. Screenshot from VF35 DTC Ranges Spreadsheet

CAN Bus	LIN Master Module	LIN 1	LIN 2	LIN 3	LIN 4	LIN 5
Body	APM	U1200-U121F				
Body	BCM	U1220-U122F	U1230-U123F	U1240-U124F		
Info	CCU	U1250-U125F				
Info	HCM	U1260-U126F				
PTCAN	ADAS	U1270-U127F	U1280-U128F			
PTCAN	EMS	U1290-U129F				
PTCAN	VCU	U12A0-U12AF				
AltCAN	XGW	U12B0-U12BF	U12C0-U12CF			

Figure 9. Screenshot from VF35 DTC Ranges Spreadsheet for LIN Master DTC Ranges

7.3 General Rules for U-Codes

- At a minimum, the ECU shall supervise the fastest periodic highest priority CAN message received from another ECU according to the CAN matrix.
- For Loss Of Communication DTC's, the Rx ECU can only report a DTC against ECU's that reside on the same CAN network.
- In the case where the Rx CAN message originates from an ECU on another CAN network, the Rx ECU shall report a Loss Of Communications with the Gateway (XGW), and the Gateway (XGW) shall report the Loss Of Communications DTC against the Tx ECU on the other CAN



bus

- The Receiving ECU must supervise at least 1 message from each source ECU (OR'd) in its DTC Test Failed logic, and with individual message timeouts.
See below example

[illegible]

Figure 10. Example of U0146-87 DTC Failure Type Definition for multiple message supervision

- If a Rx CAN message has an Alive Counter and Checksum signals, this allows End To End Protection of that message from the source to the destination ECU's, even if they reside on different CAN networks.
- When an Alive Counter or Checksum failure is detected, the Rx ECU shall report the DTC in the following form (Uxxxx.YY, where YY is either FTB=82 or 83)
- For Rx signals with states \$1=Invalid/\$0=Valid or \$0=Valid/\$1=Invalid (see Figure 4 example below from Info_CAN), if signal is received with "Invalid" state, the Rx ECU shall stop using the payload signal (since no longer valid), and report DTC UxxxxYY (where YY is FTB=81). In the case of VehicleSpdSts = \$1 (Invalid) all Receivers listed shall report DTC Uxxxx81 (Invalid Data Received From ECU xxxx) and stop using payload signal VehicleSpd until VehicleSpdSts = \$0 (Valid)
 - Transmitter Module must also report a DTC to identify the root cause of the "Invalid" state

Bus	Signal Name	Message Name	Message ID (hex)	Value Matrix	Sender	Receiver
Info-CAN	ADAS_TSR_TSPd1	ADAS_TSR	0x129	n.a	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSPdVal1	ADAS_TSR	0x129	0 "Valid" 1 "Invalid"	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSPd2	ADAS_TSR	0x129	n.a	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSPdVal2	ADAS_TSR	0x129	0 "Valid" 1 "Invalid"	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSPd3	ADAS_TSR	0x129	n.a	ADAS_Info	MHU, HUD
Info-CAN	ADAS_TSR_TSPdVal3	ADAS_TSR	0x129	0 "Valid" 1 "Invalid"	ADAS_Info	MHU, HUD
Info-CAN	MCPressure	ESC_STATUS	0x20D	n.a	XGW_Info	MHU, HUD
Info-CAN	MCPressureSts	ESC_STATUS	0x20D	1 "Invalid" 0 "Valid"	XGW_Info	MHU, HUD
Info-CAN	SAS_SteerWheelAngle	0x17E	n.a	n.a	XGW_Info	MHU
Info-CAN	SAS_SteeringAngleValid	SAS_Sensor	0x17E	1 "Valid" 0 "Invalid"	XGW_Info	MHU
Info-CAN	SAS_SteerWhlRotSpd	SAS_Sensor	0x17E	n.a	XGW_Info	MHU
Info-CAN	SAS_SteerWhlRotSpdStatus	SAS_Sensor	0x17E	1 "Valid" 0 "Invalid"	XGW_Info	MHU
Info-CAN	VehicleSpd	ESC_STATUS	0x20D	n.a	XGW_Info	MHU, HUD, CPD
Info-CAN	VehicleSpdSts	ESC_STATUS	0x20D	1 "Invalid" 0 "Valid"	XGW_Info	MHU, HUD, CPD

Figure 11. Extract from VF35 CAN Matrix (Info-CAN)

- Where ECU is receiving multiple messages with Rolling Count/Checksum or Validity signals, the DTC failure criteria shall "OR" the messages into a signal DTC. No need to create independent DTC's for each message failure. See examples below



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DTC Number (hex)	DTC Number	Failure Type Byte and meaning	DTC Name	DTC Failure Type Definition(s) (if necessary)	Repair Action	Test Failed criteria/ Confirmation Failure criteria	Mature Time	Test Pass Criteria
C43381	U043381	01 - Invalid Serial Data Received	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Invalid Serial Data Received	ESC message received with INVALID signal state	Check the relevant ECU for performance DTC	ESC message 0x732 is received with signal ABS_LFVehDataValid = 00 (invalid), or ABS_SFVehDataValid = 00 (invalid), or ESC message 0x733 is received with signal ABS_LRVehDataValid = 00 (invalid), or ABS_SRVehDataValid = 00 (invalid), or ABS_VehDataValid = 00 (invalid), or ESC message 0x734 is received with signal ESC_LateralACC_Valid = 00 (invalid), or ESC_LongitudeACC_Valid = 00 (invalid), or ESC_VehDataValid = 00 (invalid)	Immediate	ESC message 0x732 is received with signal ABS_LFVehDataValid = 01 (valid), and ABS_SFVehDataValid = 01 (valid), and ESC message 0x733 is received with signal ABS_LRVehDataValid = 01 (valid), and ABS_SRVehDataValid = 01 (valid), and ABS_VehDataValid = 01 (valid), and ESC message 0x734 is received with signal ESC_LateralACC_Valid = 01 (valid), and ESC_LongitudeACC_Valid = 01 (valid), and ESC_VehDataValid = 01 (valid)
C43782	U043782	02 - Abs / Sequence Counter Increased / Not Updated	Invalid Data Received From Information Center "X" - Abs / Sequence Counter Increased / Not Updated	Message received from SBH without the corresponding rolling count value being properly updated	Check the relevant ECU	Abs counter is not updated for message MHU_STATUS_Ba432, and MHU_FOTA_STATUS_Ba432	Immediate	Abs counter is properly updated for message MHU_STATUS_Ba432, and MHU_FOTA_STATUS_Ba432
C43783	U043783	03 - Value of Signal Protection Calculation Increased	Invalid Data Received From Information Center "X" - Value of Signal Protection Calculation Increased	Message received from SBH with an incorrect checksum calculation	Check the relevant ECU	Checksum of message MHU_STATUS_Ba432, and MHU_FOTA_STATUS_Ba432 is incorrect	Immediate	Checksum of message MHU_STATUS_Ba432, and MHU_FOTA_STATUS_Ba432 is correctly received

Figure 12. Example strategy for Invalid Data from MHU, monitoring multiple messages same source



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- When Assigning Loss Of Communication/Invalid Data Received from ECU “XXX”/Bus OFF DTC’s shall refer to figure 6 so consistency can be maintained for all ECU’s
- ECU’s shall support Bus OFF DTC for each CAN network they are connected too.

VF module name	SAE J2012 module name	Lost comms DT	Invalid data DT	Comment
APM	Anti-Pinch Module	U1004	U1004	
CCU	Climate Control Unit	U1003	U0424	
PSDM-D	“Seat Control Module A”	U0208	U0509	
BCM	Body Control Module	U0140	U0422	
ACM / ACU	Airbag Control Module	U1006	U0452	VFc35 = ACM, VFc34 = ACU
ETG	Rear Gate Module	U0230	U0531	
TRM	Trailer Module	U100C	U100C	
SHVU-F	“Seat Control Module C”	U0210	U0511	Reserving “seat control module b” for passenger seat memory module
SHVU-R	“Seat Control Module D”	U0211	U0522	
OCS	Restraints Occupant Classification System	U0154	U0455	
IPC	Instrument Panel Cluster (IPC) Control Module	U0155	U0423	
YGV	Serial Data Gateway Module “A”	U0146	U0447	
BMS	Battery Energy Control Module “A”	U0111	U0412	
AVAS	Audible Alert Control Module	U0160	U0461	
OBC	Battery Charger Control Module “A”	U0195	U043C	
DCDC	DC/DC Converter Control Module “A”	U0238	U0539	
VCU	Vehicle Dynamics Control Module	U0122	U0416	Not using “Hybrid/EV Powertrain Control Module” (U0283/U0534)
EDS-F	Drive Motor Control Module A	U0110	U0411	
EDS-R	Drive Motor Control Module B	U0232	U0535	
ESC	Anti-Lock Brakes System (ABS) Control Module	U0121	U0415	
EPS	Power Steering Control Module “A”	U0131	U0420	
ADAS	Automated Driving System Control Module “A”	U015C	U045D	
CPD	Child Presence Detection Module	U100E	U100E	
HUD	Head Up Display	U0158	U0453	
MHU	Media Head Unit	U0156	U0457	
MDU	Media Display Unit	U0257	U0558	
EMS	ECM/PCM “A”	U0100	U0401	
TCU	TCM	U0101	U0402	
GS / EKS	Transmission Range Control Module	U019E	U043F	VFc35 = GS, VFc34 = EKS
RCU	Redundant Control Unit	U110D	U100D	
IDB	Intelligent Distribution Brake	U110E	U100E	
PDU	Power Distribution Unit - OBC + DCDC	U110F	U100F	VFc35 only
NFS	Multi-Function Switches	U1130	U1030	
DW/S	Driver Window Switch	U1221		
LN2-SRGM	Sunroof Control Module	U0163		
LN1-EAS	Battery Monitor Module	U01B0		
LN2-VLC	Wireless Accessory Charging Module “A”	U020C		
LN1-SCM	Steering Column Control Module	U0212		
LN1-RLS	Rain Sensing Module	U0231		
LN2-SRBM	Sunroof Blind Module	U1231		
LN2-FKS	Foot Kick Sensor	U1232		
LN3-ALM1	LIN ambient lighting module 1	U1241		
LN3-ALM2	LIN ambient lighting module 2	U1242		
LN3-ALM3	LIN ambient lighting module 3	U1243		
LN3-ALM4	LIN ambient lighting module 4	U1244		
LN3-ALM5	LIN ambient lighting module 5	U1245		
LN3-ALM6	LIN ambient lighting module 6	U1246		
Bus OFF DTC’s	DTC Name			
	Chassis CAN	U1111		Failure Type Byte 88
	Powertrain CAN	U1112		Failure Type Byte 88
	Info CAN	U1113		Failure Type Byte 88
	Body CAN	U1114		Failure Type Byte 88
	Diag CAN	U1115		Failure Type Byte 88
	Fluxury Bus Fault	U1116		Failure Type Byte 88
	FA-CAN	U1117		Failure Type Byte 88
	BA-CAN	U1118		Failure Type Byte 88
Battery High/Low DT	DTC Name			
	Power Supply - Circuit Voltage Above Threshold	U1100		Failure Type Byte 17
	Power Supply - Circuit Voltage Below Threshold	U1101		Failure Type Byte 16
Coding DTC’s	DTC Name			
	Online Data String (ODS) - Not Configured	U1500		Failure Type Byte 55 (Not Configured) or 56 (Invalid / Incompatible Configuration)
	Factory Vehicle Order Data (VOD)	U1501		Failure Type Byte 55 (Not Configured) or 56 (Invalid / Incompatible Configuration)
	Current Vehicle Order Data (VOD)	U1502		
	Current Vehicle Order Data (VOD) Mismatch with VOD Master	U1503		
VIN DTC’s	DTC Name			
	Vehicle Identification Number - Not Configured	U3002		Failure Type Byte 55 (Not Configured)
				Yellow highlighting indicates DTC not aligned to SAE J2012

Figure 13. Communication DTC list

Contact EE-Diagnostic Engineer, or Aftersales Diagnostic and Programming team if you require DTC not listed above



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7.4 DTC Names

- If using a DTC Number from the J2012DA list, then the DTC name shall match J2012DA
- If defining a DTC from the Manufacturer Controlled Range, please ensure the DTC name is intuitive
 - Please keep the customer in mind (the Service Technician)
 - We should avoid acronym's in DTC Name unless they are obvious industry standards
 - Refer to the Vinfast Diagnostics team for guidance where unsure
- Where defining a single DTC with multiple FTB's, the DTC name shall be same (see example in Figure 7 below). The FTB will differentiate the failure mode of the DTC

DTC Number (hex)	DTC Number	Failure Type Byte and meaning	DTC Name
970D15	B170D15	0x15: Circuit short to battery or open	Air Distribution Feedback Signal
970D11	B170D11	0x11: Circuit short to ground	Air Distribution Feedback Signal
970D10	B170D10	0x10: Air Distribution Pressure Meter Calibration Error	Air Distribution Feedback Signal

Figure 7. See DTC B170D, with unique FTB

Configuration Related DTC's to consider (examples)

- VIN not Written or VIN Invalid
- Coding not written or Coding Invalid
- Key Fobs not Programmed
- Immobilizer Key Not Learned
- Steering Angle Sensor not Learned
- Airbag Module not Armed
- ADAS Camera/Radars not learned



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Assigning DID Numbers

- For ECU DID's (those not defined in the Basic DID Definition Specification), the ISO-14229 Annex C table C.1 shall be referenced for permitted ranges
- Supplier DID's shall be defined from System Supplier Specific Ranges below.

Supplier DID Range	Comments
0xFD00-0xFEFF	This range of values shall be used to reference system supplier specific record data identifiers and input/output identifiers within the server.

Figure 8. System Supplier Specific DID Range

Assigning RID Numbers

- For ECU specific RID's (those not defined in VINFAST Specifications), ISO-14229 Annex F table F.1 shall be referenced for permitted ranges
- Typically Supplier RID's will be defined from *System Supplier Specific Range* below

0x0200 - 0xDFFF	vehicleManufacturerSpecific This range of values is reserved for vehicle manufacturer specific use.	U	VMS_
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Figure 9. System Supplier Specific RID Range